

Recursive Information–Container Dynamics: A Multiscale Theory of Adaptive Systems

Computational Edition — RICS and FDFM Layers Fully Instantiated

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Editorial note on this revision. This edition completes the transition from a domain-independent theoretical architecture to a fully computable one. Three earlier corrections are retained without change: the information-manifold scope conditions in Part 4, the corrected resource-debt equation in Part 7, and the disambiguated notation for the curvature tensor, production rate, and divergence field. Beyond those, every quantity in the FDFM layer (Part 6) and the RICS/recursive layer (Parts 5, 8, 10) that was previously an existence claim — “there exists a functional $\Phi/G/\Psi/\Theta$ satisfying properties X” — has been given a concrete default formula, so that the document can be translated directly into code without a separate specification pass. Where a companion document (a Feedback Divergence Field Model application to the U.S. justice system) had already worked out concrete, computable forms for the single-container layer, those forms are adopted here as the defaults, generalized to RICD’s three-channel (official/operational/meta) representation structure and its more general recursive and network layers. A new Part 15 consolidates every concrete formula and the adapter contract a domain implementation must satisfy, as a single computational reference. Part 12’s theorems have been revisited in light of the new concrete couplings: 12.8, previously downgraded to a conjecture for lack of a specified dependence between network coherence and adaptive dimensionality, is restored to a derivable proposition now that §10.3 gives d_A an explicit formula depending on fragmentation. §10.5a gives RICD a single checkable collapse-onset condition, the direct generalization of the hull-breach condition from the companion FDFM document. **§10.5b has been corrected in this edition after further development of the underlying mechanism.** The prior draft attributed the low-divergence-apparent regime’s two possible outcomes to a coherence/residual-divergence effect borrowed directly from the high-divergence-apparent regime’s mechanism. This produced an internal contradiction: the low-divergence-apparent regime’s own definition requires D_{om} to stay elevated, which the borrowed mechanism would have read as low coherence precisely in the case that should produce the *good* outcome. The corrected account separates the two regimes’ mechanisms properly: the high-divergence-apparent regime (**Flip Condition A**) retains the coherence/dissipation account, now with two named causal routes (A1, self-propelled; A2, externally triggered) converging on the same single, terminal-locked outcome. The low-divergence-apparent regime (**Flip Condition B**) instead has two genuinely distinct outcomes, determined by fluid-mass weighting rather than coherence: **B1** (meta converges with the minority official layer) locks at 180° ; **B2** (meta converges with the majority operational layer) completes a full reorganization, since the same external multi-knock launch is carrying proportionally more of the container’s mass. New subsections on prevention-versus-post-flip asymmetry (the two regimes differ sharply in how foreseeable

their tipping events are) and steadying-versus-one-time-reset (high-divergence-apparent containers require sustained containment, not a single corrective action) have also been added, along with a note on combining anchoring with active messaging change.

Correction to the previous review (retained). The prior review flagged the codomain of the feedback permeability operator (§6.5, Definition 6.3; §11.3) as truncated in the source. On inspection of the original document, both instances are intact and correctly state $\Pi_f: \mathcal{D} \rightarrow [0,1]$; the truncation existed only in an earlier PDF-text-extraction pass, not in the source document. No correction to §6.5 or §11.3 was needed on that account.

On free parameters. Every concrete formula introduced in this edition carries one or more coefficients (α , β , κ , λ , θ , and similar) that are not fixed by the theory and are not meant to be guessed. They are exactly the analog of a , b , c , d , e , r in the companion FDFM document — free parameters with priors, to be fit to real data through the Bayesian state-space layer once a system is chosen. Part 15 tags every such coefficient explicitly so the eventual Python implementation knows which quantities are structural (fixed by the formula) and which are inferred (fit per application).

Part 1. Introduction

Abstract

Adaptive systems—from single cells to software architectures, corporations, ecosystems, and nation-states—must continuously acquire, process, transform, and act upon information while operating under finite energetic and structural constraints. Existing mathematical frameworks typically emphasize one aspect of this process, such as dynamical systems, network topology, control theory, information theory, or stochastic processes, but rarely provide a unified formalism capable of describing how information, structure, resources, and feedback jointly govern adaptive behavior across multiple organizational scales.

This paper introduces Recursive Information–Container Dynamics (RICD), a general mathematical framework for describing adaptive systems as recursively embedded information-processing containers. Within this framework, every adaptive entity is modeled as a bounded container that receives information from its environment, transforms that information according to internal organizational constraints, exchanges resources across its boundaries, and recursively participates within higher-order containers. Information, energetic cost, structural organization, and network topology are therefore treated as complementary components of a single multiscale dynamical system rather than as independent modeling assumptions.

A central component of the framework is Feedback Divergence Field Mechanics (FDFM), which formalizes the internal informational dynamics of an adaptive container. FDFM describes how discrepancies between declared, operational, and self-modeling informational states evolve through recursive feedback, generating measurable quantities

such as informational divergence, permeability, integrity, and accumulated informational stress. By treating FDFM as one layer within a broader recursive architecture, the framework separates internal informational dynamics from higher-level structural, energetic, and network processes while preserving their mathematical coupling.

Rather than postulating collapse as a threshold crossing or externally imposed state transition, RICD derives adaptive stability, degradation, and collapse from the evolution of recursive information flow under finite resource constraints. Stable adaptation emerges when recursive feedback maintains informational coherence across scales; instability arises when divergence, structural constraints, energetic depletion, and recursive memory jointly destabilize the system's attractor landscape.

The framework is intentionally domain-independent. The same mathematical objects describe biological organisms, software systems, financial institutions, educational organizations, distributed artificial intelligence, and other adaptive systems by varying only boundary definitions, resource mappings, observational operators, and recursive depth. This universality permits theoretical analysis while simultaneously supporting computational implementations such as forensic auditing systems, adaptive collapse trackers, cyber-defense architectures, organizational diagnostics, and multiscale simulation engines.

The remainder of this manuscript develops the framework axiomatically. Primitive mathematical objects are introduced first, followed by the recursive operators that govern information exchange, energetic constraints, adaptive networks, and memory. Feedback Divergence Field Mechanics is then derived as the internal informational layer of the theory, after which measurable observables, stability conditions, and collapse dynamics are developed from first principles rather than introduced as independent assumptions.

1. Introduction

Adaptive systems exist because they continuously solve a common problem: maintaining organized behavior while processing uncertain information in changing environments. Whether the system is a biological organism responding to external stimuli, a corporation coordinating thousands of employees, a software platform managing distributed computation, or a governmental institution making policy decisions, its long-term persistence depends upon the recursive interaction between information processing, structural organization, energetic constraints, and environmental feedback.

Although numerous mathematical disciplines study portions of this problem, existing approaches typically isolate one level of organization. Dynamical systems theory characterizes trajectories in state space but generally assumes a fixed state representation. Network science models interactions among agents but often treats the informational content of those interactions as secondary. Information theory quantifies uncertainty and communication while remaining largely agnostic regarding organizational structure. Control theory emphasizes regulation and stability but generally presumes externally specified objectives and architectures. Statistical mechanics explains emergent collective behavior yet does not explicitly distinguish informational organization from energetic exchange.

Adaptive systems, however, require all of these processes simultaneously. Information must exist within organizational boundaries; boundaries constrain information processing; processing consumes resources; resource availability alters organizational structure; structural change modifies future information flow; and the resulting dynamics recursively propagate across multiple organizational scales. No single mathematical framework presently formalizes these relationships as different aspects of one recursive process.

Recursive Information–Container Dynamics (RICD) is proposed as a general mathematical framework intended to address this gap. Rather than beginning with predefined state variables or application-specific differential equations, the theory begins by defining the primitive mathematical objects shared by all adaptive systems. These objects describe information, containers, boundaries, resources, recursive composition, feedback, and memory independently of any specific domain. Observable quantities such as informational divergence, permeability, rigidity, fragmentation, resilience, and collapse are subsequently derived from the interactions among these primitive objects rather than postulated independently.

Within this framework, Feedback Divergence Field Mechanics (FDFM) occupies a specific theoretical role. FDFM is not synonymous with the complete theory. Instead, it describes the internal informational dynamics occurring within individual adaptive containers. It provides a mathematical description of how official representations, operational behavior, and recursive self-models evolve through feedback, producing measurable informational integrity and divergence that influence—but do not fully determine—the larger multiscale dynamics of the system.

This layered architecture offers two complementary advantages. Theoretically, it separates informational dynamics from energetic, structural, and network processes, allowing each to be developed independently before being mathematically coupled. Practically, it permits individual layers—particularly FDFM—to be implemented as standalone computational tools for applications such as organizational diagnostics, cybersecurity monitoring, software integrity auditing, and collapse prediction without requiring the full recursive formalism.

The objective of this paper is therefore not merely to construct another model of adaptive collapse, but to establish a mathematical language capable of describing adaptive systems across domains and organizational scales. The remaining sections introduce the primitive mathematical objects, state the foundational axioms, develop the recursive operators governing adaptive dynamics, derive FDFM as the internal informational layer of the framework, and demonstrate how observable quantities associated with adaptation, resilience, and collapse emerge naturally from these first principles.

Part 2. Primitive Mathematical Objects

Recursive Information–Container Dynamics (RICD) is founded upon a small collection of primitive mathematical objects from which all subsequent structures are derived. These

primitives are not intended to represent measurable variables themselves; rather, they define the ontology of adaptive systems independently of any specific application domain. Observable quantities such as informational divergence, permeability, rigidity, adaptive capacity, energetic reserve, and collapse are introduced only after these foundational objects have been established.

The central design principle is that every adaptive system can be understood as a recursively organized collection of bounded information-processing entities interacting across multiple spatial, temporal, and organizational scales.

2.1 Information States

The fundamental object of the theory is an information state.

Definition 2.1 (Information State). An information state is represented by a probability measure:

$$P \in \mathcal{P}(\Omega),$$

where:

- Ω denotes the system's state space, and
- $\mathcal{P}(\Omega)$ denotes the space of probability measures on that state space.

An information state therefore represents not a single condition of the system, but its entire probabilistic organization. This formulation accommodates deterministic systems as degenerate probability distributions while naturally extending to uncertain, stochastic, and partially observable systems.

Unlike classical state-space models, RICD treats uncertainty as fundamental rather than incidental. Every adaptive system possesses incomplete information regarding itself, its environment, or both. Consequently, probability measures become primitive objects rather than secondary statistical descriptions.

2.2 Containers

Information cannot exist independently of organizational structure.

Definition 2.2 (Container). A container is an ordered tuple:

$$C = (\Omega, P, B, R, M)$$

consisting of:

- an information state space Ω ,
- an information state P ,
- a boundary operator B ,
- a resource state R ,
- a memory operator M .

The container constitutes the minimal adaptive entity within the theory.

The term container is deliberately chosen rather than agent or node. A container is not assumed to possess agency, intentionality, or cognition. It is simply any bounded system capable of maintaining and updating information through interaction with its environment.

2.3 Boundaries

Containers are distinguished by their boundaries.

Definition 2.3 (Boundary Operator). A boundary operator:

$$B : C \rightarrow \partial C$$

defines the permissible exchange of information, resources, and influence between a container and its environment. Importantly, the boundary is not merely geometric. It is an operator that determines admissible interactions. Two containers occupying identical physical space may possess entirely different informational boundaries.

Boundary operators therefore become dynamic mathematical objects rather than fixed spatial partitions. Their evolution will later determine permeability, organizational restructuring, and adaptive coupling.

2.4 Resources

Adaptive computation incurs finite cost.

Definition 2.4 (Resource Measure). Each container possesses a nonnegative resource measure:

$$R : C \rightarrow \mathbb{R}_{\geq 0}$$

representing the available capacity for information processing, structural maintenance, adaptation, and environmental interaction.

No assumption is made regarding the particular conservation law governing the resource.

2.5 Memory

Adaptive systems possess history.

Definition 2.5 (Memory Operator). Each container possesses a memory operator:

$$M_t : H_t \rightarrow \mathcal{A}(\Omega),$$

where:

$$H_t = \{P(s) : 0 \leq s \leq t\}$$

denotes the complete historical trajectory of the information state.

Unlike Markovian systems, future evolution depends not only upon the present state, but upon the historical trajectory encoded by M_t . Memory therefore constitutes a primitive mathematical object rather than an auxiliary delay term.

2.6 Environment

No adaptive system exists in isolation.

Definition 2.6 (Environment). The environment of a container C_i is defined as:

$$E_i = \mathcal{C} C_i,$$

where $\mathcal{C}$ denotes the collection of all containers.

The environment therefore consists of every interaction external to the container's own internal organization.

2.7 Recursive Embedding

Recursive organization is the defining characteristic of complex adaptive systems.

Definition 2.7 (Recursive Embedding). Given containers $C_1, \dots, C_n$, there exists an aggregation operator:

$$A : \{C_1, \dots, C_n\} \rightarrow C^*$$

mapping collections of lower-level containers into higher-order containers.

The resulting container C^* inherits its own information state, boundary, resource state, and memory, while simultaneously containing the original lower-level systems. Recursion therefore generates organizational hierarchy without requiring distinct mathematical rules at each scale.

2.8 Observation

Observable variables are not primitive. They arise through observation.

Definition 2.8 (Observation Operator). An observation operator:

$$O : C \rightarrow X$$

maps internal container states onto measurable quantities.

Different observers may employ different observation operators while referring to the same underlying adaptive system.

2.9 Principle of Ontological Economy

The framework deliberately minimizes primitive assumptions. Only six primitive mathematical objects are assumed:

1. information states;
2. containers;

3. boundary operators;
4. resource measures;
5. memory operators;
6. recursive aggregation.

All remaining quantities—including informational divergence, permeability, rigidity, fragmentation, adaptive capacity, resilience, collapse, and Feedback Divergence Field Mechanics itself—will be derived from these primitives in subsequent sections.

Part 3. Foundational Axioms

The primitive objects introduced in the previous section define the mathematical universe of Recursive Information–Container Dynamics (RICD). The present section specifies the fundamental principles governing those objects.

Axiom 1. Information Requires a Container. For all P , there exists a C such that $P \in C$. Information does not float freely through the theory; every probability distribution belongs to some adaptive organization. There is no such thing as a disembodied adaptive information state.

Axiom 2. Every Container Possesses Finite Capacity. For every container: $0 \leq R(C) < \infty$. Resource capacity is always finite. Every adaptive decision incurs opportunity cost.

Axiom 3. Information Processing Has Nonzero Cost. Let $P_t \rightarrow P_{t+\delta t}$ denote an informational update. Then $\text{Cost}(P_t, P_{t+\delta t}) > 0$ whenever $P_t \neq P_{t+\delta t}$. Information dynamics and energetic dynamics cannot be treated independently.

Axiom 4. Boundaries Regulate Exchange. No interaction occurs independently of boundary operators. For every interaction $\Phi : C_i \rightarrow C_j$, there exists a boundary operator B such that $\Phi = B(\Phi)$. Boundary evolution becomes a primary mechanism of adaptation.

Axiom 5. Adaptive Systems Are Recursive. Every adaptive container may itself become a component of a larger adaptive container: $A : \{C_1, \dots, C_n\} \rightarrow C$. *The resulting container C satisfies the same axioms as every lower-level container.* One mathematical language describes every organizational scale.

Axiom 6. Memory Influences Future Dynamics. Adaptive systems are generally non-Markovian: $P(t + \delta t) = F(H_t)$, where $H_t = \{P(s) : 0 \leq s \leq t\}$. Path dependence becomes fundamental, not exceptional.

Axiom 7. Adaptive Stability Is Dynamic. Adaptive systems seek bounded trajectories maintaining organizational integrity despite continual perturbation, not static equilibrium. Let $X(t)$ denote the complete state of a container. Adaptive stability requires $X(t) \in \mathcal{S}$ for some invariant stability manifold $\mathcal{S}$, rather than convergence toward a fixed equilibrium point. Collapse should later be defined as loss of stable trajectory, not departure from equilibrium.

Axiom 8. Observable Quantities Are Emergent. For any observable quantity Y , there exists an observation operator O such that $Y = O(C)$.

The Principle of Recursive Universality. Any adaptive system can be represented as a recursively embedded collection of bounded information-processing containers whose evolution is governed by finite resources, boundary-mediated exchange, historical memory, and recursive organization.

Part 4. Information Geometry

The primitive object introduced in Part 2 was the information state, represented as a probability measure over a state space. The present section equips the collection of information states with mathematical structure.

Rather than representing adaptive systems by vectors in Euclidean state space, Recursive Information-Container Dynamics models adaptive organization as evolution on a manifold of probability distributions. This choice reflects the fact that adaptive systems rarely possess complete certainty regarding either their internal organization or their environment. Their dynamics therefore consist not only of changes in state, but of changes in belief, uncertainty, expectation, and informational organization.

4.1 Information Manifold

Let Ω denote the measurable state space associated with a container, and $\mathcal{A}(\Omega)$ the space of probability measures on Ω .

Definition 4.1 (Information Manifold). An information manifold is a differentiable manifold $\mathcal{M} = (\mathcal{A}(\Omega), g)$, where g is a Riemannian metric defined on the tangent space of probability distributions.

Scope condition. $\mathcal{A}(\Omega)$ does not automatically carry a differentiable structure for arbitrary Ω . Definition 4.1 applies under one of the following restrictions, which must be specified per application:

- (i) Ω is finite or countable and $\mathcal{A}(\Omega)$ is restricted to distributions with full or fixed support, so that $\mathcal{A}(\Omega)$ is realized as an open simplex — a finite-dimensional differentiable manifold in the ordinary sense;
- (ii) the container's admissible states are restricted to a finite-dimensional parametric family $\{P_\theta : \theta \in \Theta \subseteq \mathbb{R}^n\}$ (e.g., an exponential family), so that $\mathcal{M}$ is the parameter manifold Θ pulled back through $\theta \mapsto P_\theta$, with g the induced Fisher information metric; or
- (iii) Ω is infinite-dimensional and $\mathcal{A}(\Omega)$ is treated as a Fréchet manifold, in which case the differentiability, completeness, and curvature constructions of §4.4–§4.6 require the additional analytic hypotheses standard to that setting (e.g., a fixed reference

measure and an L^2 or Orlicz-type tangent space) and are not assumed to hold without further argument.

Every subsequent construction in this section — velocity, acceleration, curvature, and geodesics — presupposes one of these restrictions has been fixed for the container in question. Each point on the manifold corresponds to an entire informational organization rather than to a single deterministic configuration. Movement on the manifold represents learning, adaptation, forgetting, reorganization, or environmental change.

4.2 Informational Distance

To compare informational organizations, the manifold requires a notion of distance. Rather than prescribing one universal metric, Recursive Information-Container Dynamics introduces an abstract informational distance functional.

Definition 4.2 (Informational Distance Functional). An informational distance is a mapping $D : \mathcal{A}(\Omega) \times \mathcal{A}(\Omega) \rightarrow \mathbb{R}_{\geq 0}$ satisfying:

1. **Non-negativity:** $D(P, Q) \geq 0$.
2. **Identity:** $D(P, P) = 0$.
3. **Continuity:** small perturbations in probability distributions produce correspondingly small changes in informational distance.
4. **Monotonicity under information loss:** coarse-graining or irreversible compression cannot increase distinguishability.

Depending upon the application, suitable choices include Fisher–Rao geodesic distance, Jensen–Shannon divergence, Wasserstein distance, Hellinger distance, total variation distance, and symmetrized Kullback–Leibler divergence.

Relationship between D and g . These choices are not interchangeable once the manifold structure of §4.1 is put to use. The scalar quantities of Part 6 (pairwise divergence, informational integrity, informational stress) require only an admissible D in the sense of Definition 4.2, and any of the six listed functionals may be used there. The Riemannian constructions of §4.4–§4.6 (velocity norm, covariant acceleration, curvature, geodesics), by contrast, are defined with respect to the specific metric g of Definition 4.1, and only Fisher–Rao distance is the geodesic distance generated by that metric on a statistical manifold; the induced local quadratic approximation of Fisher–Rao (equivalently, one-quarter the squared Hellinger distance in the appropriate limit) is the metric g used throughout §4.4–§4.6 unless otherwise stated. Wasserstein distance corresponds to a different Riemannian structure (the Otto metric on the space of measures) with its own, distinct geodesics and curvature, and total variation and symmetrized KL are not themselves Riemannian distances at all — KL is not symmetric or a metric, and TV is not induced by a Riemannian metric in the relevant sense. An application choosing Wasserstein or TV as its working D must either restrict Part 4’s differential-geometric machinery to the Fisher–Rao case for a separate purpose, or explicitly substitute the Otto geometry (with its own geodesic and curvature definitions) throughout §4.4–§4.6; the theory does not yield curvature or geodesics under TV or plain KL. This constraint should be stated explicitly in any application of the framework, since it is easy to read Definition 4.2 as offering six freely

interchangeable options when in fact only one is compatible with the differential geometry built on top of it.

4.3 Information Velocity

Adaptive systems do not merely occupy informational states. They move through informational space.

Definition 4.3 (Information Velocity). Let $P(t)$ be a trajectory on the information manifold. The instantaneous information velocity is:

$$V(t) = dP/dt.$$

Its norm, $\|V(t)\|_g$, measures the instantaneous rate of informational reorganization.

High informational velocity may represent rapid learning, crisis response, organizational restructuring, software patch deployment, or emergency adaptation. Low informational velocity represents informational stability. High velocity is neither inherently beneficial nor detrimental; its effect depends upon subsequent resource availability and structural organization.

4.4 Information Acceleration

Adaptive systems may themselves change the rate at which they reorganize.

Definition 4.4. Information acceleration is $A(t) = \nabla_t V(t)$, where ∇_t denotes the covariant derivative induced by the manifold geometry g of Definition 4.1.

Acceleration measures changes in adaptive responsiveness rather than changes in state alone.

4.5 Information Curvature

Adaptive landscapes possess geometry. Some informational regions are smooth. Others contain ridges, valleys, or unstable transition surfaces.

Definition 4.5. The curvature tensor associated with the information manifold, denoted $\mathcal{R}_\mathcal{M}$ (distinguished here from the resource measure R of §2.4), characterizes local geometric distortion under the metric g fixed in §4.1–4.2. Regions of high curvature correspond to informational organizations in which small perturbations may produce disproportionately large adaptive consequences.

Examples include tipping points, institutional crises, software instability, ecological thresholds, and cascading failures. Curvature therefore provides a geometric precursor to collapse before collapse itself is formally defined.

4.6 Information Geodesics

Adaptive systems generally attempt to reorganize efficiently.

Definition 4.6. A geodesic $\gamma(t)$ is a trajectory satisfying $\nabla_{\gamma} \dot{\gamma} = 0$ with respect to g . Geodesics represent locally minimal informational reorganization subject to the manifold geometry.

Geodesics should not be interpreted as optimal behavior. Rather, they describe informational trajectories requiring minimal intrinsic restructuring. External constraints introduced later may prevent systems from following geodesic paths.

4.7 Information Fields

Individual containers rarely evolve independently. The information state therefore becomes a field over the collection of containers.

Let $\mathcal{C} = \{C_i\}$ denote the recursive collection of containers. Define $P_i(t)$ as the informational state associated with container C_i . The resulting assignment $C_i \mapsto P_i(t)$ constitutes an information field distributed across the recursive architecture. Local informational change therefore possesses both spatial and recursive structure. This observation motivates the later development of Feedback Divergence Field Mechanics.

4.8 Information Coherence

Not every collection of informational states forms a coherent organization.

Definition 4.7 (Information Coherence). Given a collection of containers $\{C_1, \dots, C_n\}$, define a coherence functional:

$$\mathcal{C}_{\text{info}} = \Phi(P_1, \dots, P_n), \text{ where } 0 \leq \mathcal{C}_{\text{info}} \leq 1.$$

High coherence indicates mutually compatible informational organization across containers; low coherence indicates fragmentation or contradiction. The precise form of Φ is intentionally left unspecified at this stage and will later be constrained by the recursive coupling operators developed in subsequent sections.

4.9 Why Geometry Comes Before Dynamics

A common approach in adaptive modeling is to begin with differential equations and subsequently define quantities such as divergence or stability. Recursive Information–Container Dynamics adopts the opposite approach: the geometry of informational organization determines which trajectories, divergences, and transformations are mathematically meaningful. Only after the manifold structure has been established — and, per §4.1–4.2, only within the scope conditions and single-metric commitment specified there — can the dynamical equations governing adaptation be derived.

Transition to Part 5

The information manifold developed in this section describes the geometry of individual informational states. Adaptive systems, however, are rarely isolated. They are recursively nested within larger organizations while simultaneously containing lower-level

subsystems. The next section introduces the recursive operators that couple informational manifolds across scales.

Part 5. Recursive Containers

Adaptive systems exist within nested organizational structures. Cells compose tissues, tissues compose organs, organs compose organisms, organisms compose populations, and populations compose ecosystems. Similarly, software modules compose applications, applications compose distributed services, and services compose technological infrastructures.

Recursive Information–Container Dynamics treats this nested organization as a mathematical property rather than a modeling convenience. Every container may simultaneously function as an autonomous adaptive system, a component of a larger adaptive system, and an environment for lower-level adaptive systems. The same mathematical rules apply at every level of recursion.

5.1 Recursive Levels

Let C^k denote the collection of containers at recursion level k . Examples: neurons $\rightarrow$ neural assemblies $\rightarrow$ cortical regions $\rightarrow$ whole brain; or programmers $\rightarrow$ software teams $\rightarrow$ engineering divisions $\rightarrow$ corporation. No level possesses privileged mathematical status.

5.2 Aggregation Operator

Higher-order containers emerge through aggregation.

Definition 5.1. $A_k : C^k \rightarrow C^{k+1}$. For $\{C_1, \dots, C_n\}$, $C^* = A_k(C_1, \dots, C_n)$.

C^* is not equal to the sum of C_i . Aggregation is not simple addition; the resulting container possesses new organizational structure.

Emergence Principle. The aggregation operator generates new state variables that do not exist within individual components — institutional culture, organizational policy, software architecture, ecological niche, corporate governance, social norms. These quantities belong to C^{k+1} , not to any individual C^k . Emergence therefore arises mathematically rather than metaphorically.

Computational instantiation (concrete form of A). Child containers generally do not share a state space ($\Omega_1 \neq \Omega_2 \neq \dots$ in general — courts, prisons, and enforcement bureaus are the standing example), so A cannot be defined as direct pooling of the P_i . A is instantiated in two steps:

1. **Commensuration.** Each child container supplies a summary map $\sigma_i : \mathcal{R}(\Omega_i) \rightarrow \Delta^m$, projecting its own outcome distribution onto a fixed, shared m -category severity/impact scale common to the whole recursive system (e.g., a normalized harm or deviation-from-baseline scale). σ_i is domain-specified per container — this

is adapter-layer work, not engine work — but its codomain Δ^m (the m -category probability simplex) must be the same for every child of a given parent.

2. **Resource-weighted pooling.** With commensurated distributions $\sigma_i(P_i)$ now comparable, define $P^* = \sum_i [R_i / \sum_j R_j] \cdot \sigma_i(P_i)$, a resource-weighted mixture over the shared simplex Δ^m . This is the default instantiation of A_k ; it is a genuine aggregation rather than the sum of P_i (satisfying the Emergence Principle above only in the weak sense of producing a new distribution not equal to any component — full emergent structure, e.g. correlations induced by cross-container coupling, requires the network layer of Part 8 to also feed into P^* and is addressed there).

This default can be overridden by a domain-specific A_k where warranted, but P^* must remain a genuine probability distribution over some stated Ω^* for every subsequent RICD construction (D_{ij} , I , $S(t)$, etc.) to apply to C^* unchanged.

5.3 Projection Operator

Higher-level organization constrains lower-level behavior: $\Pi_k : C^{(k+1)} \rightarrow C^{(k)}$. Examples include corporate policy affecting employees, immune regulation affecting cells, operating systems constraining software processes, and constitutions constraining governmental agencies. Aggregation and projection form complementary recursive operators.

5.4 Recursive Consistency

Definition 5.2. A recursive hierarchy satisfies consistency when $O(A(C_i)) \approx \Gamma(O(C_i))$, where Γ is an admissible aggregation rule defined on observations. Perfect equality is neither expected nor desirable; emergent systems necessarily generate new organizational information.

5.5 Boundary Nesting

Correction. The earlier “If $C_i \subseteq C_j$, then $B_i \leq B_j$ ” was stated in set-containment language that doesn’t fit aggregation-type recursion: courts, prisons, and bureaus are not subsets of “the federal justice system” — C^* is an aggregate of siblings, not a superset containing them (§5.2). Set-containment nesting applies only to strict-subset recursion (e.g., district courts nested inside a court system that is itself one of the aggregated children). For aggregation-type recursion, the correct relation is compatibility, not containment: define $B_i \sqsubseteq B^*$ (“ B_i is boundary-compatible with the parent”) to mean every interaction admitted by B_i is also admissible under B^* ’s exchange rules with C_i ’s environment — i.e., the parent’s boundary operator does not forbid anything the child’s boundary operator permits toward the parent. This does not imply identical permeability in either recursion type — a department may possess more permeable informational boundaries than the corporation that contains it, and a sibling container may possess more permeable boundaries than the aggregate.

5.6 Resource Flow Across Levels

Let $R^{\wedge}(k)$ denote resource allocation at level k , and T_R the recursive transfer operator: $R^{\wedge}(k+1) = T_R(R^{\wedge}(k))$. Higher-order systems redistribute resources downward through $T_R^{\wedge}\{-1\}$. Examples include budgets, nutrients, computational allocation, attention, labor, and bandwidth.

Normalization requirement. $\sum_i R_i^{\wedge}(k)$ in the conservation principle below presupposes R_i is expressed in a common unit across all children — false in general for aggregation-type recursion (judicial docket-hours, prison bed-years, and bureau budget dollars have no natural common unit). Before summing, each child's resource measure must pass through a normalization operator $N_{R,i} : \mathbb{R}_{\geq 0} \rightarrow [0,1]$, typically utilization-based — $N_{R,i}(R_i) = R_i / R_i^{\wedge \max}$, i.e., fraction of that container's own maximum historical or designed capacity currently available — so that the conservation principle below is applied to normalized capacity shares, not raw incommensurable quantities.

Conservation Principle. $\sum_i N_{R,i}(R_i^{\wedge}(k)) + L^{\wedge}(k) = R^{\wedge}(k+1)$ (in normalized units), where $L^{\wedge}(k)$ represents transfer losses, including entropy, inefficiency, communication overhead, corruption, and leakage. Recursion naturally introduces dissipative effects. If a genuinely common unit exists for a given application (e.g., a single agency's sub-budgets, all in dollars), $N_{R,i}$ may be taken as the identity and the original unnormalized form is recovered as a special case.

5.7 Recursive Information Fields

Each recursion level supports its own information field $P^{\wedge}(k)$. Together these form $\{P^{\wedge}(0), P^{\wedge}(1), \dots\}$. An employee, a department, and an entire corporation all possess legitimate information states; none is reducible to another.

5.8 Cross-Scale Coupling

$\Lambda_{kl} : C^{\wedge}(k) \rightarrow C^{\wedge}(l)$ describes how disturbances propagate between scales. Examples: a local software bug becoming a system outage, employee burnout becoming organizational decline, species extinction altering ecosystems. $\Lambda_{kl} \neq \Lambda_{lk}$ in general — upward and downward influence generally differ, so recursive dynamics are asymmetric.

5.9 Fixed Points of Recursion

A recursive fixed point satisfies $A \circ \Pi(C) = C$. Such systems reproduce their own organizational structure across recursive scales — stable institutions, mature ecosystems, robust software architectures, healthy biological development. Collapse corresponds not merely to informational divergence, but to the destruction of recursive fixed-point structure.

5.10 The Recursive Principle

Adaptive complexity arises not from the existence of many interacting components, but from the continual bidirectional coupling between organizational levels through recursive

aggregation, projection, and resource exchange. Information moves upward. Constraints move downward. Resources move in both directions. Memory propagates across levels.

Transition to Part 6

With the recursive architecture now established, we are ready to introduce Feedback Divergence Field Mechanics (FDFM) as the internal informational layer operating within every container.

Part 6. Feedback Divergence Field Mechanics (FDFM)

FDFM concerns only the evolution of informational representations within a single bounded container. It describes how an adaptive system constructs internal representations, compares those representations with observed reality, updates them through feedback, and accumulates informational stress when those updates become progressively impaired.

6.1 Internal Information State

Let $P(t) \in \mathcal{A}(\Omega)$ denote the intrinsic informational organization of a container. This state is not directly observable. Instead, every observable informational quantity is generated from $P(t)$ through projection operators.

Definition 6.1 (Representation Operators). Define three projection operators O_o, O_p, O_m , producing $P_o = O_o(P)$, $P_p = O_p(P)$, $P_m = O_m(P)$ — interpreted as official representation, operational representation, and meta-representation, respectively, all derived from the same underlying informational state.

An organization does not possess three independent minds. Rather, it possesses one informational organization observed through three different functional lenses.

6.2 Representation Consistency

If every representation accurately reflects the underlying information state, then $P_o = P_p = P_m = P$ — maximal informational integrity. More generally, representation error is unavoidable; adaptive systems continuously approximate, rather than perfectly reproduce, their own informational organization.

6.3 Informational Divergence

Definition 6.2 (Pairwise Divergence). For any two representations P_i, P_j , define $D_{ij} = D(P_i, P_j)$, where D is an admissible informational distance from Definition 4.2 (any of the six listed functionals is valid here — this is a scalar use, not a Riemannian one; see §4.2). The pairwise divergences become D_{op}, D_{om}, D_{pm} .

These quantities measure different informational failures. Official-versus-operational divergence may reflect implementation failure, whereas operational-versus-meta divergence reflects inaccurate internal self-monitoring.

Computational instantiation (concrete default for D). For discrete or discretized Ω — the normal case for real data — take D as symmetrized, additively smoothed Jensen–Shannon divergence:

$$D(P_i, P_j) = \frac{1}{2} \cdot \text{KL}(P_i \parallel M) + \frac{1}{2} \cdot \text{KL}(P_j \parallel M), \text{ where } M = \frac{1}{2}(P_i + P_j),$$

with each empirical distribution first smoothed as $P(x) = (\text{count}(x) + \epsilon) / (N + \epsilon|\Omega|)$ for small $\epsilon > 0$ (Laplace smoothing), which keeps every KL term finite even when a category is unobserved in one representation but not the other — a routine occurrence in sparse real-world data and a hard failure mode for unsmoothed KL. This single default replaces the six-option menu of §4.2 for every scalar (non-Riemannian) use of D in Parts 6, 7, and 11; an application may substitute Wasserstein or another admissible distance from §4.2 if its Ω is naturally continuous or ordinal, but JS-with-smoothing is the default assumed by every other formula in this Part unless stated otherwise.

6.4 Feedback Operator

Define the feedback operator $F : P \rightarrow P$, incorporating environmental observations, recursive memory, internal prediction, and resource constraints, aimed at reducing expected future informational error. F need not decrease divergence at every update — temporary increases may occur during learning, organizational restructuring, crisis response, or innovation.

6.5 Feedback Permeability

Not every informational discrepancy successfully modifies the underlying information state. Some discrepancies are ignored, suppressed, forgotten, or prevented from propagating.

Definition 6.3 (Permeability Operator). Define $\Pi_f : D_{ij} \mapsto [0,1]$. For any informational discrepancy, Π_f denotes the proportion successfully transmitted into adaptive updating. $\Pi_f = 1$ represents complete informational openness; $\Pi_f = 0$ corresponds to complete informational closure.

Permeability is not a probability of communication. It is an operator governing how much informational discrepancy actually influences subsequent adaptation.

Computational instantiation (concrete default for Π_f). This is the direct RICD analog of feedback permeability as already operationalized in the companion FDFM justice-system document, generalized from a single scalar to a per-discrepancy operator. Given observable correction-and-response data for the container, define a raw permeability score

$$r_f(t) = [\text{corrective actions}(t) / \text{error signals}(t)] - \text{response lag}(t) \text{ (lag measured in the container's natural time unit, e.g. years to policy change),}$$

then map it into the required $[0,1]$ codomain via a logistic squashing function,

$$\Pi_f(t) = 1 / (1 + \exp(-\kappa_\Pi \cdot r_f(t))),$$

with κ_Π a fitted scale parameter (Part 15). This is domain-general: “corrective actions,” “error signals,” and “response lag” are placeholders the adapter fills in per container (audit recommendations implemented / issued / time-to-implement, for the justice-system case), but the aggregation formula itself does not change across domains.

6.6 Informational Integrity

Definition 6.4. Let $I = G(D_{op}, D_{om}, D_{pm})$, where $G : \mathbb{R}^3_{\geq 0} \rightarrow [0,1]$ is a decreasing functional satisfying $I = 1$ only when all pairwise divergences vanish.

Informational integrity summarizes the internal coherence of the representation system. Alignment becomes one particular manifestation of informational integrity rather than an independent primitive.

Computational instantiation (concrete default for G). $I(t) = \exp[-\alpha \cdot (D_{op}(t) + D_{om}(t) + D_{pm}(t))]$, $\alpha > 0$ fitted. This satisfies Definition 6.4 exactly: $I \in (0,1]$ always, $I = 1$ iff all three pairwise divergences vanish, and I is strictly decreasing in each divergence — so Proposition 12.3 (Bounded Informational Integrity) holds by construction rather than by separate proof for this instantiation. This is RICD’s three-channel generalization of the companion document’s messaging-divergence term inside T ; where that document collapses $O/P/\mu$ mismatch into a single scalar M before combining it with S and F , this formula keeps the three pairwise divergences distinguishable, which is a genuine strength of RICD’s representation (§6.3’s interpretation of D_{op} , D_{om} , D_{pm} as diagnostically different failure modes) worth preserving rather than flattening to match the simpler document.

6.7 Informational Stress

Definition 6.5. Informational stress is a path-dependent functional:

$$S(t) = \int_0^t K(t, s) (1 - \Pi_f(s)) D(s) ds,$$

where $K(t, s)$ is a memory kernel.

Stress accumulates whenever informational discrepancies persist while adaptive feedback remains partially blocked. This naturally incorporates technical debt, institutional denial, chronic trauma, unresolved software defects, and bureaucratic drift without ad hoc latent variables.

Computational instantiation (concrete default for K, with a recursive update). Take $K(t,s) = \lambda \cdot \exp[-\lambda(t-s)]$, $\lambda > 0$ fitted (exponential memory, the standard choice and the only one of §9.3’s candidates that admits an $O(1)$ -per-step update rather than requiring the full history to be re-integrated at every timestep). With $D(t)$ taken as the aggregate discrepancy $D(t) = D_{op}(t) + D_{om}(t) + D_{pm}(t)$ and a fixed simulation step Δt , this reduces to the recursion

$$S(t+\Delta t) = S(t) \cdot \exp(-\lambda \Delta t) + \lambda \Delta t \cdot (1 - \Pi_f(t)) \cdot D(t),$$

which is what should actually be implemented — it is exact for the exponential kernel, requires storing only the previous $S(t)$ rather than the full trajectory, and is a strict finite-difference discretization of Definition 6.5, not an approximation of a different model.

6.8 Divergence Field

Let x index informational location within the container. Define $\mathcal{D}(x, t)$ — using a distinct script symbol from the pairwise divergence D_{ij} and the general distance functional D , since all three now appear in the same coupled expressions in later sections. The resulting divergence field describes the spatial distribution of informational inconsistency. Regions of concentrated divergence represent localized adaptive failures; diffuse divergence indicates systemic degradation. This spatial formulation gives Feedback Divergence Field Mechanics its name.

6.9 Informational Equilibrium

The natural equilibrium of FDFM is not $D = 0$. Instead, adaptive informational equilibrium corresponds to bounded divergence satisfying $D(t) < D_c$, for some stability region D_c determined jointly by resource availability, feedback permeability, and recursive organization. Adaptive systems therefore tolerate limited informational inconsistency while maintaining global coherence. §10.5a gives this boundary a concrete, checkable form — the collapse inequality — once resource and permeability terms are also instantiated in Part 7.

6.10 Role Within RICD

RICD provides recursive organization, geometry, resources, memory, and network dynamics. FDFM provides informational representations, informational divergence, permeability, informational integrity, and accumulated informational stress. Together these layers describe both the structural organization of adaptive systems and the informational processes operating within them.

Transition to Part 7

The next section introduces Resource Dynamics, deriving resource expenditure directly from informational transformation.

Part 7. Resource Dynamics

Every adaptive process requires the transformation of information. Learning, memory formation, structural repair, communication, prediction, error correction, and organizational reconfiguration all require finite physical or abstract resources. Information dynamics cannot be separated from resource dynamics.

7.1 Resource Functional

Each container possesses a finite adaptive capacity represented by a resource functional $R[C] \geq 0$. Depending on the application, this may correspond to metabolic energy,

computational capacity, available capital, organizational slack, cognitive attention, time, available labor, or communication bandwidth.

7.2 Informational Work

Let $P(t) \rightarrow P(t + \delta t)$ represent an informational update. Define the informational work functional $W = W(P(t), P(t + \delta t))$. The work required depends upon the geometric displacement between informational organizations (measured by D , per §4.2's scope conditions).

7.3 Minimal Adaptive Cost

From Axiom 3, nontrivial informational updates possess strictly positive cost. There exists a lower bound $W \geq \Psi(D(P_t, P_{t+\delta t}))$, where Ψ is a monotonically increasing cost functional.

Computational instantiation (concrete default for W , satisfying Theorem 12.4's hypothesis by construction). Take $\Psi(D) = \kappa_W \cdot D^p$, $\kappa_W > 0$, $p \geq 1$ fitted, giving a baseline work requirement $W_{\text{raw}}(t) = \kappa_W \cdot D(t)^p$. To satisfy Theorem 12.4's resource-feasibility hypothesis ($W \leq w(R)$ with $w(0) = 0$) rather than merely hoping it holds, clip the work actually performed to what the container can currently fund:

$$W(t) = \min[W_{\text{raw}}(t), R(t)],$$

i.e., $w(R) = R$ itself — the simplest function satisfying $w(0) = 0$ — meaning a container facing an informational update that would cost more than its current reserve either performs a partial update (proportionally scaled-down informational change) or defers the excess to the next timestep, but never runs R below zero from this mechanism. This clipping rule is what should actually be implemented; Theorem 12.4 is the proof that doing so keeps $R(t) \geq 0$.

7.4 Resource Balance

Resources change through production, expenditure, and exchange:

$$dR/dt = P_{\text{prod}} - W + E_{\text{exch}},$$

where P_{prod} denotes productive generation, W denotes informational work, and E_{exch} denotes net exchange across the container boundary. Adaptive capacity increases through production, decreases through informational work, and is modified by interaction with the environment.

7.5 Maintenance Cost

Not all resource expenditure produces adaptation. Define $M(C)$ as the maintenance functional. Then $W = W_{\text{adaptive}} + M(C)$. Maintenance includes structural preservation, communication, monitoring, error checking, and routine coordination.

7.6 Efficiency

Define adaptive efficiency $\eta = D(P_t, P_{\{t+\delta t\}}) / W$. Large values indicate highly efficient adaptation; low values indicate resource-intensive adaptation.

7.7 Resource Resilience

Define the recovery operator Γ_R . Following disturbance, $R \rightarrow \Gamma_R(R)$. The speed and completeness of this recovery determine adaptive persistence under repeated perturbation — biological healing, financial recovery, replenishment of computational resources, organizational recruitment, psychological restoration.

Computational instantiation (concrete default for Γ_R). Simple exponential relaxation toward a domain-specified target capacity R_{target} : $dR_{\text{recovery}}/dt = \rho \cdot (R_{\text{target}} - R)$, $\rho > 0$ fitted (the recovery-rate analog of a half-life). Combined with §7.4's balance equation, the full resource update used in implementation is $dR/dt = P_{\text{prod}} - W(t) + E_{\text{exch}} + \rho \cdot (R_{\text{target}} - R)$, with $W(t)$ taken from §7.3's clipped form.

7.8 Recursive Resource Flow

$T_R^+ : R^k \rightarrow R^{k+1}$ and $T_R^- : R^{k+1} \rightarrow R^k$ describe upward aggregation and downward allocation of adaptive capacity. Recursive transfer need not be symmetric; higher-level organizations may extract resources from lower levels faster than they replenish them, or reinvest surplus capacity downward.

7.9 Resource Debt

When informational work exceeds sustainable net inflow over extended periods, adaptive systems accumulate resource debt. Consistent with the balance equation of §7.4, define:

$$\delta R(t) = \int_0^t (W - P_{\text{prod}} - E_{\text{exch}}) ds.$$

Note on this correction. The exchange term E_{exch} must appear here with the same sign convention as in §7.4, or the debt integral does not track the deficit implied by the balance equation — a container could run a genuine resource deficit masked by positive exchange, or show apparent debt that is actually being offset by exchange in real time. If a particular application intends debt to measure production shortfall net of exchange (i.e., what the container generates versus what it spends, treating exchange as a separate ledger), that must be stated explicitly and the two quantities — production-net debt and full balance-equation debt — should be reported separately, since they answer different questions. Positive δR represents cumulative adaptive deficit under the full-balance convention used here. Examples include technical debt, institutional burnout, ecological overshoot, chronic physiological exhaustion, and infrastructure neglect. This quantity is distinct from informational stress $S(t)$, though the two are mathematically coupled (§7.11).

7.10 Resource Stability

Resource stability requires:

$$\limsup_{T \rightarrow \infty} \{ (1/T) \int_0^T (P_{\text{prod}} - W) dt \geq 0,$$

using P_{prod} consistently (as defined in §7.4) rather than the bare information-state symbol P . This condition does not require continuous resource growth — only that long-term adaptive expenditure remain sustainable. Persistent violation necessarily leads to progressive degradation regardless of informational coherence.

7.11 Coupling to FDFM

Informational divergence increases the informational work required for successful adaptation. Limited resources restrict the system's ability to reduce informational divergence. This mutual dependence establishes a positive feedback loop: greater divergence $\rightarrow$ greater informational work $\rightarrow$ resource depletion $\rightarrow$ reduced adaptive capacity $\rightarrow$ greater divergence.

The Resource Principle. Resources are conserved adaptive capacity consumed through informational transformation, replenished through production and exchange, and distributed recursively across organizational scales.

Transition to Part 8

The next section develops Adaptive Network Dynamics, treating network topology as a dynamical object.

Part 8. Adaptive Network Dynamics

Adaptive systems do not evolve in isolation. Every container exchanges information, resources, and influence with other containers through continually evolving patterns of interaction. RICD treats network topology as an emergent property of adaptive dynamics rather than a fixed modeling assumption.

8.1 Interaction Graph

$G(t) = (\mathcal{C}, E(t))$, where $E(t)$ is the time-dependent set of interactions. Topology evolves continuously alongside the adaptive system.

8.2 Weighted Interactions

$W_{ij} = (I_{ij}, R_{ij}, P_{ij})$, where I_{ij} represents informational exchange, R_{ij} represents resource exchange, and P_{ij} represents effective permeability. An interaction possesses magnitude, direction, and functional character.

8.3 Emergent Connectivity

Define the coupling functional $\Xi(C_i, C_j)$. An interaction exists whenever $\Xi(C_i, C_j) > \theta$, where θ is a domain-dependent coupling threshold. Connectivity emerges from interaction rather than being assumed beforehand.

Computational instantiation. $\Xi(C_i, C_j)$ = historical interaction frequency $\times \min(\Pi_{f,i}, \Pi_{f,j})$ — an edge is live only if both containers have historically interacted *and* neither side's permeability is near zero (a container that ignores everything cannot be meaningfully coupled to, regardless of contact frequency). θ is fitted per application.

8.4 Dynamic Topology

$dE/dt = G_{\text{gen}}(P, R, M, B)$, where the topology generator G_{gen} depends upon informational organization, available resources, historical memory, and boundary evolution.

Computational instantiation. Implemented as a per-timestep re-evaluation of §8.3's Ξ against θ over all container pairs — edges are added when Ξ crosses above θ and removed when it falls below $\theta(1 - \text{hysteresis})$, a small hysteresis band (fitted) preventing edges from flickering in and out at the threshold every step.

8.5 Information Propagation

$dP_i/dt = F_i + \sum_j [\Lambda_{ij} (P_j - P_i)]$, where Λ_{ij} is the effective coupling operator. The first term describes intrinsic adaptation; the second, network-mediated influence. Λ_{ij} may itself depend upon trust, bandwidth, boundary permeability, historical interaction, and resource availability.

Computational instantiation. $\Lambda_{ij}(t) = \gamma_{\text{net}} \cdot W_{ij}(t) \cdot \Pi_{f,i}(t)$, $\gamma_{\text{net}} > 0$ fitted — network-mediated influence scales with current edge weight and with the receiving container's own permeability (a container that doesn't absorb feedback internally won't absorb it from neighbors either, by the same mechanism). This is the direct generalization of the companion FDFM document's neighbor-misalignment update (§12.B.1 there, " S_i increases with neighbor misalignment") from a single fragmentation scalar to a full distributional update.

8.6 Resource Propagation

Let $\Phi_R(i, j)$ denote the resource flux from C_i to C_j . $\sum_j \Phi_R(i, j)$ determines the net exchange component appearing in the resource balance equation of §7.4. Information and resources propagate through distinct, though coupled, channels.

8.6a Extractive Coupling and the Parasitic Cycle

§8.6's resource flux is symmetric and neutral by default — a normal exchange channel. It is not always neutral. A container in the high-divergence-apparent regime (§10.5b) characteristically draws resource from a neighbor, fails to convert it to usable capacity, and vents the failure back onto the source as harm. This is a distinct, asymmetric coupling mode that needs its own formal treatment, because it is the actual mechanism connecting Part 7's resource layer to Part 10's tipping dynamics — not a metaphor layered on top of them.

Target selection. A high-divergence-apparent container does not extract indiscriminately from any neighbor; it preferentially targets neighbors that remain kinetically coupled after being disturbed, rather than neighbors that readily decouple. Define a coupling-persistence function $\rho(j) \in [0,1]$ for a neighbor container j , reflecting how much of a disturbance's

kinetic effect on j persists as continued oscillation (rather than being absorbed and damped): $\rho(j)$ is characteristically high for containers in the low-divergence-apparent regime (§10.5b) — their apparent instability keeps them actively rocking and coupled — and low for containers in the low-entropy baseline regime (§10.5b), which more readily returns to equilibrium and drops out of active coupling. This is why extraction characteristically targets the medium-entropy population rather than the healthy baseline: baseline containers are poor extraction targets by construction, not by choice on either side.

Extraction flux. For high-divergence-apparent container i and a coupled neighbor j :

$$X_{\{i,j\}}(t) = \chi \cdot \Lambda_{ij}(t) \cdot \rho(j) \cdot R_j(t), \chi > 0 \text{ fitted,}$$

drawing down j 's resource directly in the balance equation of §7.4 (appearing as a negative E_{exch} component for j).

Metabolic failure and heat conversion. A high-divergence-apparent container's own closed, self-propelled internal dynamics (§10.5b's F_i -dominance) mean it cannot productively convert an inflow of externally-sourced open resource into usable capacity — define a metabolic efficiency $\eta_i \in [0,1]$, characteristically near zero for high-divergence-apparent containers, giving a heat term:

$$H_i(t) = (1 - \eta_i) \cdot X_{\{i,j\}}(t),$$

representing the fraction of extracted resource that cannot be metabolized and must be discharged rather than retained.

Venting and the origin of the medium-entropy driving force. $H_i(t)$ is not dissipated harmlessly; it is vented back into the network as additional forcing, directed at j if j remains coupled ($\rho(j)$ still high), or redistributed across other available coupled neighbors if j has decoupled. This vented heat is not a separate force from §10.5b's $\sum_j \Lambda_{ij}(t)$ driving term for medium-entropy tipping — **it is that term's actual origin**: the “lateral knocks” driving a medium-entropy container toward its own tipping event are, mechanistically, the discharged metabolic failure of the high-divergence-apparent containers extracting from it. §10.5b's network-mediated drive should accordingly be read as substantially composed of vented $H_i(t)$ terms from coupled high-divergence-apparent neighbors, not an independent disturbance source.

Link-severing as the high-divergence-apparent container's own collapse trigger. If j successfully transitions to the low-entropy baseline regime (§10.5b), $\rho(j)$ falls, the coupling weakens (§8.7's rewiring), and $X_{\{i,j\}} \rightarrow 0$. If i has no other available coupled target, $H_i(t)$ can no longer be vented externally — it has nowhere left to go but back into i 's own intrinsic dynamics, directly increasing the F_i term already identified in §10.5b as high-divergence-apparent tipping's driving force. This closes a mechanistic loop the theory previously described only narratively: a successfully-decoupling neighbor is not a bystander to a high-divergence-apparent container's eventual collapse, it is a direct contributor to it, and severing an extraction link is simultaneously a self-protective act for the decoupling container and a destabilizing one for the container it decouples from.

8.7 Adaptive Rewiring

$dW_{ij}/dt = H(I, \Pi_f, R, M)$, where informational integrity, feedback permeability, available resources, and historical interaction jointly determine edge evolution.

Computational instantiation (direct import from the companion FDFM document’s agent-based layer). $W_{ij}(t+\Delta t) = W_{ij}(t) + \beta_{\text{net}} \cdot (\text{successful joint updates}) - \gamma_{\text{net}} \cdot (\text{ignored feedback across the edge})$, where “successful joint update” and “ignored feedback” are counted from realized decreases versus persistent increases in $D_{ij} \cdot (1 - \Pi_f)$ across the edge over the step — this is FDFM’s §12.B.3 rule ($F_i(t+1) = F_i(t) + \beta \cdot \text{successful} - \gamma \cdot \text{ignored}$) applied to edge weight instead of node-level permeability, since RICD already has per-container Π_f and needs the analogous update for the coupling strength between containers specifically.

8.8 Cascading Dynamics

Let $\delta_i(t)$ be a localized disturbance. $\Delta(t) = L_{\text{prop}}(G(t), \delta)$, where L_{prop} is the propagation operator. Depending upon network topology, identical local disturbances may dissipate, remain localized, amplify, or cascade globally.

Computational instantiation. Implemented directly by simulating §8.5’s coupled dP_i/dt system forward from an initial perturbation at container i ; no separate propagation operator needs to be built, since Λ_{ij} already carries disturbances between containers by construction. L_{prop} is therefore not a new primitive — it is what the already-specified system does when integrated, and Part 15 treats it as such (i.e., it does not appear as a separate function in the eventual code).

§8.8a: The Cascading Grid Failure Sequence

The propagation operator defined in §8.8 describes how localized disturbances move through a network. It does not, by itself, distinguish between three qualitatively different propagation regimes: dissipation, amplification, and cascading grid failure. The distinction arises not from the topology alone, but from the *regime composition* of the containers involved and the *sequential coupling* of extraction, decoupling, venting, and tipping.

This section formalizes the specific sequence by which a single high-entropy (high-divergence-apparent) container can trigger a grid-level cascade, and the conditions under which that cascade terminates in fragmentation rather than reorganization.

8.8a.i The Cascade-Initiating Event

Let container ii be in the **high-divergence-apparent regime** (§10.5b), with:

- Low official-layer variance: $\sigma o(i) \ll \sigma p(i)$ $\sigma o(i) \ll \sigma p(i)$
- Active extractive coupling (§8.6a) to one or more neighbors $j \in Ni(t)$ $j \in Ni(t)$

- A vented heat term $H_i(t) = (1 - \eta_i) \cdot X_{i,j}(t) > 0$ for at least one coupled neighbor j

Let at least one such neighbor j decouple from i at time t_d :

- $\rho(j) \rightarrow 0$ (coupling persistence collapses)
- $X_{i,j}(t) \rightarrow 0$ for $t > t_d$
- The edge W_{ij} begins to decay via the rewiring rule in §8.7

Definition 8.8a.1 (Decoupling-Triggered Heat Accumulation). For a high-divergence-apparent container i , define the accumulated unvented heat at time t as:

$$H_i(t) = \int_{t_d}^t H_i(s) \cdot 1[N_i(s) = \emptyset] ds$$

where $1[N_i(s) = \emptyset]$ indicates that i has no remaining coupled extraction targets at time s . This quantity represents the heat that would have been vented externally but is now trapped internally.

Lemma 8.8a.1 (Internal Forcing Escalation). For a high-divergence-apparent container i , $H_i(t) > 0$ is a strictly increasing function of the duration $t - t_d$ for which $N_i(t) = \emptyset$, provided $H_i(t) > 0$. Moreover, $H_i(t)$ feeds directly into the intrinsic driving term $F_i(t)$ of §8.5:

$$F_i(t) = F_i(0) + \lambda_H \cdot H_i(t)$$

where $\lambda_H > 0$ is a fitted coupling constant. Thus, as victims decouple, the container's own internal oscillation intensifies—it is not "calming down"; it is accelerating toward its own tipping event.

8.8a.ii The Frantic Striking Phase

As $H_i(t)$ increases and $F_i(t)$ escalates, the high-divergence-apparent container enters a **frantic striking phase**.

Definition 8.8a.2 (Frantic Striking). A container i is in a frantic striking phase when:

1. $H_i(t) > H_{\text{threshold}}$ (fitted, per §15.3)
2. $N_i(t) \neq \emptyset$ but the set of coupled neighbors is decreasing
3. $\Delta_{ij}(t)$ increases in magnitude for remaining neighbors, even as $W_{ij}(t)$ decays, because the *forcing* per edge increases

This is not strategic or adaptive behavior. It is a **thermodynamic discharge**—the container is venting internally accumulated heat outward, through whatever remaining channels exist.

Proposition 8.8a.1 (Frantic Striking as Network Forcing). During a frantic striking phase, the vector sum of forcings from i_i to its remaining neighbors $N_i(t)$ is:

$$\Delta i(t) = \sum_{j \in N_i(t)} \Delta ij(t) \cdot u^{ij}(t) \quad \Delta i(t) = \sum_{j \in N_i(t)} \Delta ij(t) \cdot u^{ij}(t)$$

where $u^{ij}(t)$ is the direction of the disturbance vector from i_i to j_j . This vector sum is typically **non-zero and large**, characterized by divergent directions because i_i is striking out at multiple neighbors simultaneously—there is no single "target direction" because the heat must be discharged in all available directions.

8.8a.iii The Medium-Entropy Launch

These frantic strikes land on the remaining coupled neighbors. The containers most affected are those in the **low-divergence-apparent (medium-entropy) regime** (§10.5b), because:

- They remain coupled longer ($\rho(j)$ is high for medium-entropy containers)
- They have large operational mass ($\mu_p > \mu_o$), making them kinetically responsive to external forcing
- They are visibly chaotic at the official layer (σ high), which makes them appear as "active" participants in the dynamics, even though their operational core is calm

Definition 8.8a.3 (Medium-Entropy Launch). A medium-entropy container j_j is launched when it receives a multi-directional forcing input from multiple high-entropy neighbors:

$$\text{DriveB}(j,t) = \left| \sum_{k \in H_j(t)} \Delta kj(t) \cdot u^{kj}(t) \right| \quad \text{DriveB}(j,t) = \sum_{k \in H_j(t)} \Delta kj(t) \cdot u^{kj}(t)$$

where $H_j(t)$ is the set of high-divergence-apparent neighbors of j_j at time t . Per §10.5b, this is the vector-aware coupling—single knocks cancel, but multiple divergent knocks reinforce.

Lemma 8.8a.2 (Launch Threshold). A medium-entropy container j_j tips into a tipping event (per §10.5a) only if:

$$\text{DriveB}(j,t) > \Theta_{\text{launch}} \quad \text{DriveB}(j,t) > \Theta_{\text{launch}}$$

for some fitted threshold Θ_{launch} , and only if the multi-directional input persists over a sufficient duration to overcome the damping effect of a single source.

8.8a.iv The Bifurcation of Medium-Entropy Tipping

Once launched, the medium-entropy container follows Flip Condition B (§10.5b), which bifurcates into two distinct outcomes:

Outcome	Convergence Event	Mass Weighting	Result
B1	$\text{Dom} \rightarrow 0$	$\mu_o \mu_o$ (minority)	Terminal lock at 180° —container becomes frozen, requires external anchor
B2	$\text{Dpm} \rightarrow 0$	$\mu_p \mu_p$ (majority)	Full 360° reorganization—stress and debt reset; adaptive capacity recovers

Proposition 8.8a.2 (Cascade Propagation via B1 Locking). A medium-entropy container that undergoes B1 locking:

- Severs its remaining couplings (per §8.7) as its adaptive capacity collapses
- Ceases to be a sink for extraction heat—its $\rho(j)\rho(j)$ drops to zero
- Transitions from a coupled neighbor to a **frozen node**, which no longer participates in network dynamics

This severs one more extraction channel for the original high-entropy container, accelerating its own heat accumulation and eventual collapse.

Proposition 8.8a.3 (Cascade Damping via B2 Reorganization). A medium-entropy container that undergoes B2 reorganization:

- Resets its stress $S(t)$ and debt $\delta R(t)$ toward zero
- Recovers adaptive capacity $dA(t)$
- May re-enter the low-entropy baseline regime if the reorganization is sustained

This container ceases to be a source of lateral knocks and may become a stabilizing anchor (§10.5b) for other containers.

8.8a.v Grid-Level Fragmentation

The cascade propagates as:

- High-entropy containers decouple from their victims → strike out frantically → launch medium-entropy neighbors → those neighbors either lock (B1) or reorganize (B2)
- Locked nodes (B1) are removed from the active network; reorganized nodes (B2) may stabilize
- The network's algebraic connectivity $\lambda_2(L(t))$ declines as locked nodes accumulate

Definition 8.8a.4 (Cascading Grid Failure). Cascading grid failure occurs when:

$$\lambda_2(L(t)) \rightarrow 0 \quad \lambda_2(L(t)) \rightarrow 0$$

and the network fragments into disconnected components. This is the direct consequence of:

1. High-entropy containers losing extraction targets and self-accelerating
2. Medium-entropy containers locking (B1) faster than they reorganize (B2)
3. The network's capacity for information and resource exchange falling below the minimum required for coherence (§12.1, Criterion 12.1)

Theorem 8.8a.1 (Cascade Termination Conditions). A cascading grid failure terminates under one of the following conditions:

1. **External intervention:** An anchoring container (per §10.5b) stabilizes high-entropy containers, preventing further frantic striking.
2. **B2-dominated reorganization:** A sufficient fraction of medium-entropy containers complete B2 reorganization and re-enter the low-entropy baseline, restoring connectivity.
3. **Complete fragmentation:** The network fragments into isolated components, and the cascade stops because there are no remaining coupled neighbors to launch. This is not recovery—it is dissolution.

The conditions under which each outcome occurs are determined by the relative rates of B1 versus B2 transitions, which are governed by:

- The fraction of high-entropy to medium-entropy containers in the network
- The $\rho(j)\rho(j)$ persistence of medium-entropy containers
- The availability of external anchors
- The fitted thresholds M_{lock} and M_{Mc} in §10.5b

8.8a.vi Observational Signatures

The cascade sequence produces testable signatures that distinguish it from simple network degradation:

1. **Pre-cascade:** A high-entropy container shows $\sigma\sigma\sigma\sigma$ low, $\sigma\rho\sigma\rho$ high, and a declining number of coupled neighbors.
2. **Frantic striking onset:** Multiple medium-entropy containers show a simultaneous spike in $DriveB(j,t)$ from the same high-entropy source.
3. **Bifurcation:** Locked nodes (B1) show $Dom \rightarrow 0$, $I(t) \rightarrow 0$, and $dA(t)$ collapse; reorganized nodes (B2) show $Dpm \rightarrow 0$, $S(t)$ and $\delta R(t)$ reset.

4. **Grid-level fragmentation:** $\lambda_2(L(t))\lambda_2(L(t))$ declines monotonically until either stabilization or fragmentation.

These signatures are distinct from those of isolated container collapse (§10.5b) and can be used to detect cascading failure *in progress*, not just after the fact.

8.8a.vii Relation to Existing Sections

This section does not introduce new mathematics. It provides a formal narrative that connects:

- §8.6a (extractive coupling, heat venting, decoupling)
- §8.7 (adaptive rewiring, edge decay)
- §8.8 (propagation operator, cascading dynamics)
- §8.9 (network resilience, algebraic connectivity)
- §8.11 (network phase transitions, fragmentation)
- §10.5b (regime classification, Flip Conditions A and B, B1/B2 bifurcation)
- §11.10 (collapse potential, network coherence)

The cascade sequence is the **mechanistic link** between these previously separate layers. It is the dynamic that makes RICD a theory of *multiscale systemic collapse*, not just a collection of container-level diagnostics.

8.8a.viii Computational Instantiation

For an implementation, the cascade sequence is simulated by:

1. Tracking $H_i(t)H_i(t)$ and $N_i(t)N_i(t)$ for each high-divergence-apparent container
2. Computing $DriveB(j,t)DriveB(j,t)$ for each medium-entropy container
3. Simulating the B1/B2 bifurcation per §10.5b
4. Updating the network graph $G(t)G(t)$ and computing $\lambda_2(L(t))\lambda_2(L(t))$ at each timestep
5. Detecting cascading grid failure when $\lambda_2(L(t)) < \lambda_{threshold}\lambda_2(L(t)) < \lambda_{threshold}$ (fitted)

No new integration scheme is required. The cascade emerges from the existing coupled differential equations when the conditions above are satisfied.

End of §8.8a

Summary of Changes

Addition	Purpose
Definition of decoupling-triggered heat accumulation ($H_i(t)H_i(t)$)	Formalizes the internal build-up of vented heat when victims decouple
Lemma on internal forcing escalation	Connects $H_i(t)H_i(t)$ directly to $F_i(t)F_i(t)$
Definition of frantic striking phase	Formalizes the "striking out frantically" behavior
Vector-aware forcing sum ($\Lambda_i(t)\Lambda_i(t)$)	Explains why frantic strikes are multi-directional and constructive for medium-entropy launches
Definition of medium-entropy launch ($DriveB(j,t)DriveB(j,t)$)	Formalizes the multi-knock mechanism from §10.5b
B1/B2 bifurcation as cascade propagation/damping	Shows how locking accelerates collapse, while reorganization dampens it
Definition of cascading grid failure ($\lambda_2(L(t)) \rightarrow 0, \lambda_2(L(t)) \rightarrow 0$)	Connects container-level dynamics to grid-level fragmentation
Cascade termination conditions	Gives testable predictions for when cascades stop
Observational signatures	Makes the cascade sequence empirically distinguishable

8.9 Network Resilience

Define the network resilience functional $R_N = \Psi(G, W, P, R)$. High values indicate that information and resources continue to circulate effectively following perturbation.

Computational instantiation. $R_N(t) = \lambda_2(L(t)) \cdot \text{mean}(\Pi_f \text{ across containers})$, where $\lambda_2(L)$ is the algebraic connectivity of the network's (weighted) graph Laplacian — a standard, well-behaved connectivity measure that is zero exactly when the graph is disconnected and increases with how well-connected it is — scaled by mean permeability, since a well-connected network of containers that don't absorb feedback isn't actually resilient in RICD's sense. Network coherence N (§11.8) is set equal to a normalized $R_N \in [0,1]$ (R_N divided by its value on the complete graph with $\Pi_f = 1$, i.e., its theoretical maximum for that container count) so N and R_N are not two independently-defined quantities.

8.10 Recursive Networks

$G^*(0), G^*(1), G^*(2), \dots$ represent networks operating at different organizational scales, interacting through the recursive operators of Part 5.

8.11 Network Phase Transitions

As informational integrity declines or resources become depleted, gradual changes in local interactions may accumulate into abrupt changes in global topology — fragmentation into disconnected components, collapse of communication pathways, emergence of isolated clusters, runaway synchronization, centralized bottlenecks. These transitions arise from the coupled evolution of informational dynamics, resource constraints, and adaptive rewiring rather than from topology alone.

The Network Principle. Adaptive networks are continuously evolving organizations whose topology emerges from informational compatibility, resource exchange, historical memory, and boundary-mediated interaction.

Transition to Part 9

The next section develops Memory Dynamics.

Part 9. Memory Dynamics

Adaptive systems are fundamentally historical. RICD treats memory as an active operator that continuously modifies informational evolution, resource allocation, network topology, and recursive organization.

9.1 Historical Trajectory

$H_t = \{P(s) : 0 \leq s \leq t\}$. Future evolution depends upon H_t , not solely upon $P(t)$.

9.2 Memory Operator

Definition 9.1. $M_t : H_t \rightarrow \mathcal{H}(\Omega)$. M_t does not reproduce history exactly; it constructs an adaptive representation of history. Memory is selective, compressive, and reconstructive.

9.3 Memory Kernel

$K(t, s) \geq 0$ for $0 \leq s \leq t$. The effective remembered history becomes $M_t = \int_0^t K(t, s) P(s) ds$. Examples include exponential decay, power-law memory, oscillatory memory, finite retention windows, and multi-timescale kernels.

9.4 Memory Plasticity

$dM/dt = P_M(P, R, \Pi_f)$, where P_M denotes the plasticity operator (distinct from information state P and production rate P_{prod} — three uses of the letter P now appear across this framework and must be disambiguated by context or, in application-specific derivations, by distinct subscripts).

9.5 Memory Compression

$C_M : H_t \rightarrow \tilde{H}_t$, where $\tilde{H}_t$ is a lower-dimensional adaptive representation of the historical trajectory. Compression sacrifices detail while preserving information judged relevant for future adaptation.

9.6 Memory Reconstruction

$R_M : \tilde{H}_t \rightarrow \tilde{P}$. The reconstructed state $\tilde{P}$ may differ from the original historical state because reconstruction is influenced by current informational organization, accumulated experience, and contextual cues.

9.7 Memory and Informational Stress

$S(t) = \int_0^t K(t, s) (1 - \Pi_f(s)) D(s) ds$. Stress represents a memory-weighted accumulation of unresolved informational discrepancies, emerging from the interaction between memory and feedback dynamics rather than existing independently.

9.8 Memory and Network Evolution

$dW_{ij}/dt = H(P, R, M, \Pi_f)$. Repeated successful interaction strengthens coupling; repeated unresolved divergence weakens it.

9.9 Recursive Memory

$M^{(0)}, M^{(1)}, M^{(2)}, \dots$ represent memories operating simultaneously across organizational scales — cellular memory, individual learning, organizational culture, institutional precedent, evolutionary history.

9.10 Memory Geometry

Define a history-dependent metric $g_t = g(M_t)$. The informational distance between two states depends not only on those states themselves but also on the adaptive history through which they are evaluated. Two systems occupying identical instantaneous informational states may therefore experience fundamentally different adaptive trajectories because their historical geometries differ.

9.11 Adaptive Forgetting

Define the forgetting operator $F_M : M_t \rightarrow M_{t'}$. Adaptive forgetting removes informational detail whose maintenance costs exceed its future adaptive value. Both excessive forgetting and excessive retention can impair adaptive performance.

The Memory Principle. Memory is an active operator that compresses, reconstructs, weights, and transforms historical experience, thereby modifying the geometry of future adaptation across every recursive level of organization.

Transition to Part 10

The next section synthesizes these components into a unified theory of adaptive stability and collapse.

Part 10. Adaptive Stability and Collapse

Adaptive systems continually reorganize in response to internal dynamics and environmental perturbation. Collapse is not the opposite of stability; it is the progressive loss of the adaptive mechanisms that make stability possible.

10.1 Adaptive State Space

$X(t) = (P, R, G, M)$, consisting of informational organization, adaptive resources, interaction network, and memory. $X(t) \in \mathcal{A}$, the adaptive state manifold.

10.2 Stable Adaptive Region

$\mathcal{S} \subseteq \mathcal{A}$. A system is adaptively stable whenever $X(t) \in \mathcal{S}$. $\mathcal{S}$ is generally a dynamically evolving region of viable adaptive organization, not a point.

10.3 Adaptive Degrees of Freedom

Let $F_A(X)$ denote the set of feasible adaptive transformations available from state X . Define $d_A(X) = \dim F_A(X)$. Large values indicate flexibility; small values indicate increasing rigidity. Collapse begins when adaptive options disappear faster than they can be regenerated.

Computational instantiation (the central missing piece, now resolved). $F_A(X)$ has no domain-independent enumeration — “what transformations are available” is irreducibly domain-specific, the same way “what counts as an outcome” (Ω) is. But d_A itself does not need to be measured by directly counting transformations; it can instead be derived from quantities RICD has already made concrete: informational stress $S(t)$ (§6.7), resource debt $\delta R(t)$ (§7.9), and recursive/network fragmentation $F(t)$ (§11.7, §8.9’s N). All three are, by construction, things that consume adaptive options — accumulated unresolved stress forecloses responses that would otherwise be available, resource debt forecloses anything requiring uncommitted capacity, and fragmentation forecloses coordinated multi-container

responses. This gives a derived, computable adaptive-capacity functional rather than an independently-measured one:

$$d_A(t) = d_A^{\max} \cdot \exp[-\beta_S \cdot \tilde{S}(t) - \beta_R \cdot \delta\tilde{R}(t) - \beta_F \cdot \tilde{F}(t)],$$

where $d_A^{\max}$ is a domain-specified ceiling (the maximum plausible number of independent adaptive levers for that container type, set once per adapter — see Part 15), $\tilde{S}$, $\delta\tilde{R}$, $\tilde{F}$ are each stress/debt/fragmentation normalized to $[0,1]$ by the adapter's own historical range, and $\beta_S, \beta_R, \beta_F > 0$ are fitted weights. This is not a claim that stress, debt, and fragmentation are the only things that determine adaptive capacity — it is a claim that they are the three RICD already tracks with real formulas, so d_A can be computed without inventing a fourth independent measurement pipeline. An adapter with a genuine independent proxy for feasible options (e.g., a literal count of currently-unconstrained policy levers, as discussed for the justice-system case) may substitute it for d_A directly; the exponential-decay form above is the default when no such direct count is available.

Recursive aggregation of d_A (parent from children). Given a parent $C^* = A(C_1, \dots, C_n)$ per §5.2, define

$$d_A(C) = [\sum_i w_i \cdot d_A(C_i)] \cdot (1 - F(C)),$$

with w_i the same resource weights used in §5.2's pooling step and $F(C^*)$ the parent-level fragmentation (§11.7) between the children's projected states. This makes the parent's adaptive capacity bounded above by the resource-weighted sum of its children's capacities, discounted by how poorly those children's adaptive options actually cohere — a system whose subsystems each have many options but whose options don't compose (high fragmentation) has less real flexibility at the aggregate level than the sum suggests, which is the qualitative claim Proposition 12.8 needed and did not previously have a mechanism for (see the revised 12.8 below).

10.4 Adaptive Rigidity

$R_g = \Phi(d_A)$, where $\Phi' < 0$. Rigidity increases as adaptive degrees of freedom decrease, derived from the geometry of adaptive possibility rather than introduced independently.

Computational instantiation. $R_g(t) = 1 - d_A(t)/d_A^{\max}$, the direct normalized complement of §10.3's d_A — monotonically decreasing in d_A as required, bounded in $[0,1]$, and requiring no separate fitted parameters beyond those already used to compute d_A itself.

10.5 Collapse Operator

Definition 10.1. The collapse operator $\mathcal{C}$ maps adaptive trajectories toward lower-dimensional adaptive manifolds: $\mathcal{C}: \mathcal{P} \rightarrow \mathcal{P}$, where $\dim(\mathcal{P}) < \dim(\mathcal{N})$. Collapse is dimensional contraction within adaptive state space.

10.5a The Collapse Inequality (Hull-Breach Condition)

§6.9 left the boundary of informational equilibrium, $D(t) < D_c$, as an unspecified stability region. With I, S, Π_f , and d_A now all concrete (§6.6, §6.7, §6.5, §10.3), that boundary can

be written as a single checkable inequality — RICD’s direct generalization of the companion FDFM document’s hull-breach condition, $M + S > \theta F$ (§6 there, “divergence production exceeds feedback correction capacity”):

$$(1 - I(t)) + \tilde{S}(t) > \theta_{\text{hull}} \cdot \Pi_f(t)$$

where $\tilde{S}(t)$ is stress normalized to $[0,1]$ by its own historical range (matching §10.3’s normalization) and $\theta_{\text{hull}} > 0$ is a fitted threshold. The left side is total unresolved divergence pressure — informational incoherence plus accumulated stress; the right side is correction capacity — feedback permeability scaled by how much correction the container’s threshold tolerates. The inequality holding at time t is a collapse-onset signal: divergence-and-stress pressure has outrun the container’s demonstrated capacity to correct it. This is a genuine generalization rather than a coincidental resemblance — setting $I \approx 1 - M$ (treating messaging divergence as the complement of integrity) and $\Pi_f \approx F$ recovers the companion document’s original condition as a special case where S is dropped and I collapses to a single scalar.

§10.5a establishes *when* a container crosses from bounded oscillation into a tipping event. It does not distinguish what happens next — whether the container is a mid-transition system that will nonetheless stabilize, a resource-locked system that briefly de-stabilizes without permanent damage, or a genuinely different transformation entirely. §10.5b develops that distinction: the same threshold crossing has at least three qualitatively different resolutions, determined by which channels converge, what force drives the event, and how much of that force survives as usable momentum through the transition.

10.5b Entropy Regime Classification and the Convergent Collapse/Reset Mechanism

Hull breach identifies *when* a container tips. This section identifies what determines the outcome once it does, why formally similar convergence events between representation channels produce opposite outcomes depending on regime, how a container arrives in a given regime in the first place, and what interventions are and are not safe. Every quantity below is built from primitives already defined elsewhere in this document, including §8.6a’s extractive-coupling mechanism, which turns out to be load-bearing rather than incidental to what follows.

Entropy regime classification. Let $\sigma_o(t)$, $\sigma_p(t)$ denote the trailing-window variance of P_o and P_p respectively — both already computable from the adapter’s raw O_o , O_p feeds (§15.2), no new observation required. Classify a container into one of three regimes, not two:

- **Low-entropy baseline regime:** D_{op} , D_{om} , D_{pm} all small; Π_f high (near 1, the meta-layer is genuinely bidirectionally permeable); σ_o and σ_p both low-to-moderate — a slight, rhythmic rocking rather than violent oscillation. Critically, D_{om} is not required to be zero, and should not be: a small nonzero D_{om} — the official layer under mild, healthy self-questioning rather than perfect, unexamined confidence — functions as a negative-feedback channel that prevents the container from developing an unchecked, grandiose official narrative. $D_{om} \in (0, \varepsilon_{\text{healthy}}]$ is the target band, not $D_{om} = 0$. This regime is the reference state the

other two are diagnosed as deviations from, and is what a healthy container occupies by default absent a genesis-branching event, defined next.

- **High-divergence-apparent regime:** $\sigma_o(t)$ low, $\sigma_p(t)$ high — the official/declared layer presents as stable while the operational layer is actually turbulent underneath it.
- **Low-divergence-apparent regime:** $\sigma_o(t)$ high, $\sigma_p(t)$ low — the official layer presents as visibly unstable while the operational layer is actually steady beneath it — the mirror case.

These are not different containers in kind; the same container can occupy any of the three at different points in its trajectory, and classification is a diagnostic read on the current relationship between apparent and actual volatility, not a fixed type.

Regime genesis: developmental branching. A container does not begin in either non-baseline regime; it arrives there through a shock event acting on a previously low-entropy container. Let $S_{\text{shock}}(t)$ denote a large, discrete, exogenous perturbation to $D(t)$ or $R(t)$ — structurally, an extraction event per §8.6a, or an equivalent one-off environmental disturbance. Define an ambient verticality-signal $V(t) \in [0,1]$, sourced either directly from the container that inflicted S_{shock} (transmitted during the shock itself) or diffusely from the surrounding network's regime composition (a stochastic, grid-level field — in human systems, the ambient promotion of entitlement or domination scripts). Branching at a shock event resolves as:

- **$V(t) > v_{\text{threshold}}$:** the operational layer absorbs the corruption — a discontinuous jump in D_{op} — and the container transitions to the high-divergence-apparent regime. The official layer remains untouched, which is precisely what allows it to continue presenting as stable while operational reality diverges underneath it.
- **$V(t) \leq v_{\text{threshold}}$:** the official layer absorbs the corruption instead — a discontinuous jump in D_{om} — and the container transitions to the low-divergence-apparent regime. The operational core remains untouched, pristine, and calm; only the declared/self-presented layer becomes visibly chaotic.

This gives the theory an account of onset it previously lacked entirely: §10.5b as first written diagnosed containers already in a non-baseline regime without explaining how they arrived there. The branch is fully determined by which layer absorbs a shock, not by the shock's magnitude alone, which is why two containers experiencing comparably severe disturbances can end up in structurally opposite regimes.

The confrontation trigger. As $\Pi_f(t)$ rises, a non-baseline container is increasingly forced to register $D_{\text{op}}(t)$ directly rather than let it pass unprocessed. Define a transient forcing term, active only in a short window once $\Pi_f(t)$ crosses a confrontation threshold Π_c while $D_{\text{op}}(t)$ remains elevated:

$$F_{\text{conf}}(t) = \zeta \cdot \Pi_f(t) \cdot D_{\text{op}}(t), \zeta > 0 \text{ fitted, nonzero only near the } \Pi_c \text{ crossing.}$$

This is a genuine self-intensification, not correction — a brief amplification of the very behavior producing the divergence, immediately following the point at which the container is made to confront it, before any resolution occurs one way or the other. It supplies the

initial push that carries a container from ordinary bounded oscillation (§6.9) into candidate tipping dynamics. A container whose Π_f never crosses Π_c , or whose D_{op} is already low, never enters a tipping event and remains in bounded oscillation regardless of what follows below. Note that a *slow* rise in Π_f — one whose rate stays below the crossing dynamics that trigger F_{conf} , rather than merely its level — can avoid activating this term altogether; this becomes the operative mechanism for the rogue-metamessenger pathway described near the end of this section.

Convergence events and their regime-dependent valence. Once a container is in a tipping event, its outcome depends on which pair of channels subsequently converges toward zero divergence — and the same convergence event carries opposite meaning depending on regime, because which layer is actually reliable flips between regimes:

- **$D_{pm} \rightarrow 0$** (meta converges with operational): in the high-divergence-apparent regime, the self-monitoring layer validates an actually-chaotic operational reality — internal correction is disabled, since Π_f 's correction mechanism depends on a detected discrepancy between P_m and P_p to act on, and none remains. In the low-divergence-apparent regime, the self-monitoring layer instead recognizes an actually-stable operational reality beneath a visibly unstable official layer — genuine reorganization around a real signal.
- **$D_{om} \rightarrow 0$** (meta converges with official): in the high-divergence-apparent regime, the self-monitoring layer holds to the declared standard against a corrupted operational layer; this does not itself prevent a tipping event already underway, but determines the container's capacity for rapid reorganization once the corrupted operational layer is stripped away post-collapse. In the low-divergence-apparent regime, the self-monitoring layer instead abandons an actually-accurate operational signal and validates an externally corrupted official narrative — internal correction is disabled in the same structural way as the high-divergence-apparent bad case, despite involving the opposite pair of channels.

Driving force and convergence pathways. The force actually driving a container through a tipping arc is already decomposed in §8.5's propagation equation, $dP_i/dt = F_i + \sum_j \Lambda_{ij}(P_j - P_i)$. The two regimes are driven by different, dominant terms in that same equation — but each regime admits more than one causal *route* to its outcome, and the low-divergence-apparent regime specifically has two genuinely distinct outcomes rather than one, determined by which channel converges:

- **Flip Condition A (high-divergence-apparent, always terminal-locked).** Two causal routes converge on the same outcome. **A1 (usual):** self-propelled — F_i -dominated, the container's own ongoing internal churning (fed, per §8.6a, by vented extraction-heat once a coupled neighbor decouples), with $D_{pm} \rightarrow 0$ convergence. **A2 (occasional):** externally triggered — an external strike combines with a $D_{om} \rightarrow 0$ collision event, producing oscillation larger than self-propulsion alone could generate; the container, careening back around, finds no counter-oscillating partner to strike against, and the unopposed momentum forces the same terminal-locked outcome as A1. Both routes end in the single outcome

defined below ($M_{\text{lock}} \leq M_{\text{tip}} < M_{\text{c}}$) — this regime does not have a 360° option under either route.

- **Flip Condition B (low-divergence-apparent, externally driven, two distinct outcomes).** Always $\sum_j \Lambda_{ij}$ -dominated — driven by simultaneous multi-directional lateral knocks from neighboring high-divergence-apparent containers, which launch the container higher, and for longer, than any self-generated oscillation could. Which of two outcomes results depends on which channel converges during that launch: **B1**, $D_{\text{om}} \rightarrow 0$ (meta converges with the visibly chaotic official layer) engages only the smaller fluid mass — official and meta, the minority layers in this regime — producing 180°, terminal-locked. **B2**, $D_{\text{pm}} \rightarrow 0$ (meta converges with the actually-calm operational layer) engages the larger fluid mass — operational and meta, since operational holds the majority of a low-divergence-apparent container's mass by the regime's own definition (the “bulk of the water,” per the originating thought experiment) — and that larger, coherently-moving mass, combined with the greater launch height shared by both B routes, completes the full 360° transition rather than stalling partway through it.

Vector-aware coupling: why a single knock damps and multiple divergent knocks propel.

§8.5's Λ_{ij} term must be treated as directional, not scalar, for either B route to behave correctly. A single dominant external source imparts a force whose direction reverses over that source's own oscillation cycle; summed over the tipping window, this is substantially self-cancelling and functions as a *restoring*, damping influence — a container knocked by one neighbor rocks and returns, rather than tips. Only *simultaneous*, *multi-directional* knocking — several high-divergence-apparent neighbors striking from divergent vectors at once — removes the restoring cancellation and produces a net constructive force:

$$\text{Drive}_B(t) = | \sum_j \Lambda_{ij}(t) \cdot \hat{u}_j(t) |,$$

where $\hat{u}_j(t)$ is the unit direction of neighbor j 's disturbance at time t . Single-source or aligned-opposing inputs partially cancel (small effective Drive_B , damping outcome, no tipping event at all); genuinely divergent multi-source inputs reinforce (large effective Drive_B , tipping event triggered, outcome then determined by B1 vs B2 below).

Fluid-mass weighting: why the same launch produces two different outcomes. Flip Condition A's two routes both terminate the same way, so the coherence-and-dissipation account below applies to A without modification. Flip Condition B's two outcomes cannot be explained the same way — the low-divergence-apparent regime's own definition requires D_{om} to be elevated, so treating D_{om} as “residual conflict left over” would produce the contradictory conclusion that the 360° outcome (B2) is *low*-coherence, which is wrong. What actually distinguishes B1 from B2 is not how much divergence is left unresolved elsewhere, but how much of the container's total representational mass is carried by whichever pair is converging. Define mass weights μ_o , μ_p , μ_m for the official, operational, and meta layers respectively (fitted per container, or adapter-supplied directly if a domain has a natural measure of relative layer salience), with $\mu_p > \mu_o$ structurally in the low-divergence-apparent regime — operational is the majority layer there by definition.

The effective driving momentum for each B route is weighted by the mass of its converging pair:

$$M_{\text{tip}}(B1) = \int \mu_o \cdot \text{Drive}_B(t) dt, M_{\text{tip}}(B2) = \int \mu_p \cdot \text{Drive}_B(t) dt,$$

drawing on the *same* shared $\text{Drive}_B(t)$ — the same external launch — but scaled by structurally different mass fractions. Because $\mu_p > \mu_o$ in this regime, B2 is structurally more likely to cross into the reorganization range defined below than B1, for the same external launching force.

Coherence, dissipation, and Flip Condition A's momentum functional. For Flip Condition A specifically, not all of a tipping event's driving force survives to complete the transition; some is dissipated as internal friction, proportional to whatever divergence remains unresolved outside the converging pair. Define the residual divergence at a convergence event as the pairwise divergence not involved in the convergence itself — for A1/A2's $D_{\text{pm}} \rightarrow 0$ or $D_{\text{om}} \rightarrow 0$ respectively, the residual is D_{op} , consistently elevated in this regime by definition, which is why this mechanism produces no contradiction here — and a coherence factor:

$$\kappa(t) = 1 / (1 + D_{\text{residual}}(t)), \kappa \in (0,1],$$

$$\text{with } M_{\text{tip}}(A) = \int |F_i(t)| \cdot \kappa(t) dt.$$

Outcome thresholds. Two thresholds, $M_{\text{lock}} < M_c$ (both fitted), apply uniformly across all routes above, with each route's own M_{tip} computed by its own mechanism (mass-weighted for B1/B2, coherence-weighted for A):

- **$M_{\text{tip}} < M_{\text{lock}}$:** no tipping event completes. The container self-arrests and returns to bounded oscillation within §6.9's equilibrium region.
- **$M_{\text{lock}} \leq M_{\text{tip}} < M_c$:** terminal-locked outcome per Definition 10.1 — reached by A1, A2, or B1. Requires an external anchor to quiet residual oscillation before recovery can begin.
- **$M_{\text{tip}} \geq M_c$:** full reorganization outcome — reached by B2 specifically, and structurally unreachable by any A route or by B1, since B1's smaller mass weighting keeps it below M_c under ordinary conditions. $S(t)$ and $\delta R(t)$ reset toward zero; $d_A(t)$ recovers toward d_A^{max} .

Prevention versus post-flip intervention: an asymmetry in predictability. Preventing a tipping event outright is preferable to managing its aftermath for both regimes, but the two regimes differ sharply in how *foreseeable* the event is, which changes where practical effort should concentrate. A low-divergence-apparent (medium-entropy) container's impending tipping event is comparatively visible in advance — its official layer is, by regime definition, the visibly chaotic one, directly observable as $\sigma_o(t)$ rising. Because its outcome genuinely could go either way (B1's lock or B2's reorganization) and is not reliably distinguishable in advance, prevention — steadying the container before it tips at all, via external anchor, rather than allowing the tip and hoping for B2 — is substantially safer than letting the event occur. A high-divergence-apparent (high-entropy) container's impending tip is, by contrast, comparatively invisible in advance, precisely because its official layer presents as

stable while the actual instability is hidden in the operational layer; this is why post-collapse intervention receives comparatively more emphasis for this regime in practice — not because prevention is less valuable in principle, but because it is harder to act on before the fact.

Steadying versus one-time reset. The two regimes also require different *kinds* of post-intervention handling, not just different timing. A low-divergence-apparent container that completes a B2 reorganization, or is steadied pre-flip, benefits from external anchoring primarily as a speed multiplier — reorganization can occur without it, but proceeds faster with it. A high-divergence-apparent container is structurally different: merely righting or resetting it without addressing internal composition is insufficient, since its unaltered internal mechanics (F_i , still driven by whatever unresolved D_{op} and unvented H_i remain) will trigger renewed violent oscillation almost immediately. What this regime requires is sustained **steadying** — continuous external containment disallowing both further oscillation and further extraction/displacement (§8.6a) — maintained over time, not applied once. Sustained steadying is also the mechanism by which a high-divergence-apparent container's composition can gradually shift toward the low-divergence-apparent or low-entropy baseline regimes, per Axiom C1: with extraction and oscillation both disallowed for long enough, F loses the external support it needs to sustain convergence around the corrupted signal, and drifts toward the true one instead.

Combined intervention. For both regimes, anchoring (a mechanical/behavioral intervention, arresting oscillation and extraction) and active messaging change (an informational intervention — external correction, or the internal rogue-metamessenger pathway below) are more effective together than either alone, and this holds whether applied pre-flip as prevention or post-flip as recovery: anchoring alone can halt an oscillation without resolving what caused it; messaging change alone, absent containment, risks triggering F_{conf} 's destabilizing surge before it can take effect. Applied together, anchoring buys the time messaging change needs to work without triggering premature confrontation.

Anchor eligibility. Not every external intervention is safe, and safety depends on the structural, not just the intentional, character of the anchor:

- A **low-entropy baseline container** can safely anchor a destabilizing neighbor, but only by actively maintaining zero extraction and zero displacement flux for the full duration of the intervention — i.e., holding $X_{\{anchor,i\}}(t) = 0$ (§8.6a) throughout, a maintained behavioral condition, not a structural guarantee. Lapsing on this vigilance risks the anchor itself being drawn into an extraction cycle and reclassified into the low-divergence-apparent regime — becoming a casualty rather than a stabilizer.
- A **semi-rigid horizontal field** — a boundary configuration with no exploitable structure to extract from or displace onto, per §2.3 — is structurally, not behaviorally, safe: $X = 0$ by construction, not by maintained effort. This is the preferred anchor wherever available, but is characteristically rare, particularly in human systems.

Rogue metamessengers and slow-correction reversal. A high-divergence-apparent container is not always healed only from outside. A small, coordinated subset of a container's own meta-representation apparatus — aware the container's operational layer is corrupted and actively working against what the container's official layer believes it is doing — can raise $\Pi_f(t)$ deliberately, provided the rate of increase, not merely its eventual level, stays below whatever threshold triggers F_{conf} 's destabilizing surge (above). Formally: a controlled injection $d\Pi_f/dt = \varepsilon_{\text{rogue}}$, with $\varepsilon_{\text{rogue}}$ small and bounded well under the rate that would cross Π_c abruptly. Held slow enough, this permits D_{op} to be metabolized incrementally through ordinary correction rather than forcing a confrontation event, allowing gradual regime transition — high-divergence-apparent toward low-divergence-apparent or, ideally, toward the low-entropy baseline — without ever triggering Flip Condition A. This pathway pairs naturally with steadying, above: sustained external containment provides the time a slow internal correction needs to work.

Two collapse-layer axioms. The mechanisms above rest on two claims general enough, and load-bearing enough, to state as axioms of the collapse layer specifically (distinct from Part 3's foundational axioms, which govern the base RICD formalism independent of any collapse dynamics):

Axiom C1 (Truth-Only Reorganization). The feedback operator F (§6.4) admits stable, unforced fixed points only at convergence toward the container's true underlying signal. Convergence toward a false signal is not a fixed point F can reach or sustain on its own; it requires continuous external or self-generated forcing (F_{conf} , or an extractive input per §8.6a) to be maintained, and decays toward the true-signal fixed point once that forcing is withdrawn. A system may reorganize around the truth of its own accord; it cannot do so around a falsehood without ongoing external support for that falsehood.

Axiom C2 (Runaway Acceleration). As a container approaches a hull-breach crossing (§10.5a) while remaining in Flip Condition A, its instantaneous divergence pressure accelerates rather than approaching the threshold at constant or decelerating rate — mass (in the bowl thought experiment's terms) gathers disproportionately just before the structural weak point is reached. This is stated as a falsifiable claim, not a narrative device: it predicts that $d(D_{\text{op}})/dt$ and dS/dt should show a measurable, characteristic upward inflection in the window immediately preceding a hull-breach crossing, distinguishable from a null model in which divergence approaches the threshold as an undifferentiated random walk. A real container's pre-collapse trajectory data can be tested against this prediction directly; the axiom is false for any domain where no such inflection is observed.

§10.5b Entropy Regime Classification and the Convergent Collapse/Reset Mechanism

Hull breach identifies when a container tips. This section identifies what determines the outcome once it does, why formally similar convergence events between representation channels produce opposite outcomes depending on regime, how a container arrives in a given regime in the first place, and what interventions are and are not safe. Every quantity below is built from primitives already defined elsewhere in this document, including

§8.6a's extractive-coupling mechanism, which turns out to be load-bearing rather than incidental to what follows.

§10.5b.0: The Topology of Verticality and Horizontality (Shaking Bowls Axioms 1 and 2)

The entropy regime classification developed in this section distinguishes containers by the relationship between their official and operational layers. That classification, however, describes *symptoms*—not the *structural origin* of those symptoms. The present subsection supplies that origin: a topological and thermodynamic account of why containers assume vertical (rigid, closed, extractive) or horizontal (fluid, open, negentropic) orientations, and why the transition between them is governed by a 90° inversion of Form over Function.

These principles are drawn from the *Shaking Bowls* thought experiment, which models containers as curved, rising walls filled with message-bearing fluid. They are not metaphors. They are **structural axioms** that govern the geometry of the container boundary and its relationship to informational dynamics.

Axiom 1: Verticality is Rigid (Form)

Informal statement:

When a fluid, open-energy exchange field rises vertically, it concentrates energy into mass, hardening into "form." High velocity or pressure enforces this vertical rigidity.

Formalization:

Let a container $C=(\Omega,P,B,R,M)$ have a **verticality measure** $h(t) \in [0,\infty)$, defined as the height of the container's boundary above its base along the axis of maximum structural extension. This is not a physical height in all domains—it is a *structural height*: the degree to which the container's boundary enforces hierarchical, closed, or impermeable constraints.

Definition 10.5b.0.1 (Verticality Operator).

Define the verticality operator $V:C \rightarrow R_{\geq 0}$ as:

$$V(t) = \Phi V(dh/dt, d^2h/dt^2, S(t), \delta R(t), \Pi f(t))$$

where:

- dh/dt is the rate of vertical boundary expansion (rigidification)
- d^2h/dt^2 is the acceleration of that expansion
- $S(t)$ is informational stress (§6.7)
- $\delta R(t)$ is resource debt (§7.9)

- $\Pi f(t)\Pi f(t)$ is feedback permeability (§6.5)

$\Phi V \Phi V$ is monotonically increasing in each of its first four arguments and monotonically *decreasing* in $\Pi f(t)\Pi f(t)$ —meaning that as permeability collapses, verticality increases. The verticality operator is normalized such that $V(t) \in [0,1] V(t) \in [0,1]$.

Axiom 1 (Formal).

For any container CC , as verticality $V(t)V(t)$ increases beyond a domain-specific threshold $\Theta V \in (0,1) \Theta V \in (0,1)$, the container undergoes a **phase transition** from an open, fluid energy exchange field to a **rigid, closed-circuit form**, characterized by:

1. $\Pi f(t) \rightarrow 0 \Pi f(t) \rightarrow 0$ (feedback permeability collapses)
2. $Dop(t) \rightarrow 1 Dop(t) \rightarrow 1$ (official-operational divergence maximizes)
3. $dA(t) \rightarrow 0 dA(t) \rightarrow 0$ (adaptive capacity contracts)
4. $RV(t) \rightarrow 1 RV(t) \rightarrow 1$ (rigidity functional reaches maximum, per §10.4)

Proposition 10.5b.0.1 (Verticality and the High-Entropy Regime).

A container in the high-divergence-apparent regime (§10.5b.1) is characterized by $V(t) > \Theta V V(t) > \Theta V$, $\Pi f(t) \ll 1 \Pi f(t) \ll 1$, and $Dop(t) \gg 0 Dop(t) \gg 0$. Its vertical rigidity is the thermodynamic origin of its self-sustaining, extractive, and blind dynamics. The container's official layer appears calm ($\sigma\sigma\sigma$ low) because the vertical walls have *frozen* the official messaging in place, while the operational layer churns chaotically beneath ($\sigma\rho\sigma$ high).

Axiom 2: Horizontality is Fluid (Function)

Informal statement:

Horizontal states represent open, uncompressed systemic flow.

Formalization:

Let a container CC have a **horizontality measure** $w(t) \in [0,\infty) w(t) \in [0,\infty)$, defined as the degree to which the container's boundary permits open, bidirectional, uncompressed exchange across its surface. This is not physical width—it is *structural width*: the degree to which the container's boundary is permeable, non-hierarchical, and fluid.

Definition 10.5b.0.2 (Horizontality Operator).

Define the horizontality operator $W:C \rightarrow R_{\geq 0} W:C \rightarrow R_{\geq 0}$ as:

$$W(t) = \Phi W(\Pi f(t), I(t), -dDop/dt, dR/dt) W(t) = \Phi W(\Pi f(t), I(t), -dtdDop, dtdR)$$

where:

- $\Pi f(t)\Pi f(t)$ is feedback permeability
- $I(t)I(t)$ is informational integrity (§6.6)
- $-dDop/dt - dtdDop$ is the rate of divergence reduction (alignment)
- $dR/dtdtdR$ is the rate of resource replenishment

$\Phi W \Phi W$ is monotonically increasing in each argument. The horizontality operator is normalized such that $W(t) \in [0,1]$.

Axiom 2 (Formal).

For any container CC , horizontality $W(t)$ is maximized when:

1. $\Pi f(t) \rightarrow 1$ (full permeability)
2. $I(t) \rightarrow 1$ (full informational integrity)
3. $Dop(t) \rightarrow 0$ (full alignment)
4. $dR/dt > 0$ (resource sustainability)

In this state, the container is an **open, uncompressed systemic flow**, characterized by:

- Bounded stress $S(t) < \Theta S$
- High adaptive capacity $dA(t) \approx dA_{max}$
- Low rigidity $RV(t) \rightarrow 0$
- Recovery time $\tau(t)$ short and bounded (§10.10)

Proposition 10.5b.0.2 (Horizontality and the Low-Entropy Baseline).

A container in the low-entropy baseline regime (§10.5b.1) is characterized by $W(t) > \Theta W$, $\Pi f(t) \approx 1$, and $Dop(t) \approx 0$, for some threshold $\Theta W \in (0,1)$. Its horizontality is the thermodynamic expression of open, negentropic, self-aware systemic flow—the state toward which all recovery trajectories ultimately converge.

The 90° Inversion Principle

The thought experiment identifies a critical isomorphism: at 90°, Form and Function invert. Verticality becomes rigidity; horizontality becomes fluidity. The same geometric configuration—a vertical line intersecting a horizontal plane—produces structural rigidity regardless of direction. This is observed in fluid dynamics: a person falling vertically onto flat water "pancakes" against a concrete-like wall of pressure, just as a vertical tsunami wave moving horizontally "pancakes" a standing observer with that same solid force. The direction of the vector does not matter; the 90° intersection of a vertical line and a horizontal plane always yields structural rigidity.

Definition 10.5b.0.3 (90° Inversion Operator).

Define the inversion operator I_{90} as a mapping from a horizontal field HH to a vertical container $CVCV$:

$$I_{90}: H \rightarrow CV$$

where:

- HH is a container with $W(t) > \Theta W$, $h(t) \approx 0$

- CVCV is a container with $V(t) > \Theta V(t) > \Theta V$, $w(t) \approx 0$, $w(t) \approx 0$

Inversion Principle.

The operator $I90I90$ is **self-dual**: applying it to a horizontal field produces a vertical container; applying it to a vertical container (via phase transition, external shock, or collapse) can produce a horizontal field—but only through **collapse** (180° flip, terminal lock) or **reorganization** (360° flip, full reset).

Mathematically:

$I90(H) = CV, I90(CV) = \{CV'(180^\circ \text{ flip: locked, inverted, still vertical})H(360^\circ \text{ flip: full reorganization to horizontal field})$
 $I90(H) = CV, I90(CV) = \{CV'H(180^\circ \text{ flip: locked, inverted, still vertical})(360^\circ \text{ flip: full reorganization to horizontal field})$

Proposition 10.5b.0.3 (The 90° Inversion and Systemic Collapse).

A container undergoing a 180° flip (Flip Condition A, or B1) has experienced an irreversible 90° inversion of its structural orientation: it has crossed the vertical threshold and cannot return to horizontality without external anchoring. A container undergoing a 360° flip (B2) has completed *two* 90° inversions, returning to horizontality with its structure intact and its informational organization reset.

Corollary 10.5b.0.1 (Mass Weighting and the 90° Inversion).

The outcome of a 90° inversion depends on *which layer carries the majority of the container's mass*. In the high-divergence-apparent regime, the majority mass is in the operational layer, but it is *chaotic*—so a 90° inversion drives the container into a 180° lock. In the low-divergence-apparent regime, the majority mass is also in the operational layer, but it is *calm*—so a 90° inversion, when combined with sufficient external forcing, can carry the container through a second 90° inversion to a full 360° reorganization.

Ultimate Convergence: Return to the Horizontal Plane

The thought experiment observes that the ultimate evolutionary objective for all systems within the grid is the relaxation of vertical walls and the return to the foundational horizontal matrix. This is not dissolution in the sense of destruction—it is *flattening*: the container becomes less vertical, more horizontal, without losing its structural identity.

Definition 10.5b.0.4 (Ultimate Convergence).

A container CC achieves ultimate convergence when:

$$V(t) \rightarrow 0, W(t) \rightarrow 1, h(t) \rightarrow 0, B(t) \rightarrow \text{Bopen} \quad V(t) \rightarrow 0, W(t) \rightarrow 1, h(t) \rightarrow 0, B(t) \rightarrow \text{Bopen}$$

where Bopen is a boundary operator that permits full bidirectional exchange with no vertical constraint.

Proposition 10.5b.0.4 (Ultimate Convergence and Negentropy).

If a purely vertical structure represents maximum entropy and a rigid, closed circuit, then a purely horizontal plane represents maximum negentropy—an open, fluid plane of endlessly rippling, self-aware energy. Ultimate convergence is therefore a thermodynamic transition from a high-entropy, closed container to a low-entropy, open field. This is the state toward which all recovery trajectories ultimately tend, and the state from which all healthy containers originate.

Corollary 10.5b.0.2 (Horizontal Anchors and Ultimate Convergence).

Because a horizontal plane contains no vertical walls to weaponize, it acts as an absolute thermodynamic sink, absorbing expelled heat as it ripples across the energy field harmlessly (§8.6a, §10.5b.11). This is why only horizontal fields—or semi-rigid horizontal anchors—can safely stabilize high-entropy containers without being drawn into the extractive cycle.

Relationship to the Entropy Regime Classification

The entropy regime classification in §10.5b.1 can now be understood as a **diagnostic readout** of the container's underlying verticality/horizontality orientation:

Regime	Structural Signature	Verticality/Horizontality
High-divergence-apparent	$\sigma\sigma\sigma\sigma$ low, $\sigma\rho\sigma\rho$ high, $\Pi f \Pi f$ low	$V(t) > \Theta V, W(t) < \Theta W$
Low-divergence-apparent	$\sigma\sigma\sigma\sigma$ high, $\sigma\rho\sigma\rho$ low, $\Pi f \Pi f$ moderate	$V(t) < \Theta V, W(t) < \Theta W$ (partial verticality; bulk horizontality in operational layer)
Low-entropy baseline	$\sigma\sigma\sigma\sigma$ and $\sigma\rho\sigma\rho$ low-to-moderate, $\Pi f \Pi f$ high	$W(t) > \Theta W, V(t) \rightarrow 0$

This coupling is not arbitrary. The regime classification describes what is *observable*; the verticality/horizontality axioms describe what is *structurally true*.

§10.5b.1: Entropy Regime Classification

Let $\sigma_o(t)\sigma_o(t)$ and $\sigma_p(t)\sigma_p(t)$ denote the trailing-window variance of PoPo and PpPp respectively—both computable from the adapter's raw OoOo, OpOp feeds (§15.2). Classify a container into one of three regimes:

1. **Low-entropy baseline regime:** Dop, Dom, Dpm all small; Πf high (near 1); $\sigma_o\sigma_o$ and $\sigma_p\sigma_p$ both low-to-moderate—a slight, rhythmic rocking rather than violent oscillation. Dom is not required to be zero; a small nonzero Dom functions as a healthy negative-feedback channel. $Dom \in (0, \epsilon_{\text{healthy}}]$ is the target band.
2. **High-divergence-apparent regime:** $\sigma_o(t)\sigma_o(t)$ low, $\sigma_p(t)\sigma_p(t)$ high—the official layer presents as stable while the operational layer is turbulent underneath.
3. **Low-divergence-apparent regime:** $\sigma_o(t)\sigma_o(t)$ high, $\sigma_p(t)\sigma_p(t)$ low—the official layer presents as visibly unstable while the operational layer is steady beneath—the mirror case.

These are not fixed types; the same container can occupy any of the three at different points in its trajectory.

§10.5b.2: Regime Genesis (Developmental Branching)

A container arrives in a non-baseline regime through a shock event acting on a previously low-entropy container. Let $S_{\text{shock}}(t)$ denote a large, discrete, exogenous perturbation to $D(t)$ or $R(t)$. Define an ambient verticality signal $V(t) \in [0, 1]$, sourced either directly from the container that inflicted the shock or diffusely from the surrounding network's regime composition.

Branching resolves as:

- $V(t) > v_{\text{threshold}}$: the operational layer absorbs the corruption—a discontinuous jump in Dop —and the container transitions to the high-divergence-apparent regime.
- $V(t) \leq v_{\text{threshold}}$: the official layer absorbs the corruption instead—a discontinuous jump in Dom —and the container transitions to the low-divergence-apparent regime.

§10.5b.3: The Confrontation Trigger

As $\Pi f(t)$ rises, a non-baseline container is increasingly forced to register $Dop(t)$ directly. Define a transient forcing term:

$F_{\text{conf}}(t) = \zeta \cdot \Pi f(t) \cdot Dop(t)$, $\zeta > 0$ fitted, nonzero only near the Πc crossing. $F_{\text{conf}}(t) = \zeta \cdot \Pi f(t) \cdot Dop(t)$, $\zeta > 0$ fitted, nonzero only near the Πc crossing.

This is a brief self-intensification, not correction. A container whose $\Pi f \Pi f$ never crosses $\Pi c \Pi c$ or whose DopDop is already low never enters a tipping event.

A slow rise in $\Pi f \Pi f$ —one whose rate stays below the crossing dynamics that trigger FconfFconf —can avoid activating this term; this is the operative mechanism for the rogue-metamessenger pathway described below.

§10.5b.4: Convergence Events and Regime-Dependent Valence

Once in a tipping event, the outcome depends on which pair of channels converges:

- **$\text{Dpm} \rightarrow 0 \text{Dpm} \rightarrow 0$ (meta converges with operational):**
 - High-divergence-apparent: self-monitoring validates a chaotic operational reality—internal correction disabled.
 - Low-divergence-apparent: self-monitoring recognizes an actually-stable operational reality beneath a visibly unstable official layer—genuine reorganization.
- **$\text{Dom} \rightarrow 0 \text{Dom} \rightarrow 0$ (meta converges with official):**
 - High-divergence-apparent: self-monitoring holds to the declared standard against a corrupted operational layer—does not prevent tipping, but determines capacity for rapid reorganization post-collapse.
 - Low-divergence-apparent: self-monitoring abandons an actually-accurate operational signal and validates an externally corrupted official narrative—internal correction disabled.

§10.5b.5: Driving Force and Convergence Pathways

The force driving a container through a tipping arc is decomposed in §8.5's propagation equation:

$$d\Pi dt = F_i + \sum_j \Lambda_{ij} (P_j - P_i) \quad d\Pi dt = F_i + \sum_j \Lambda_{ij} (P_j - P_i)$$

Flip Condition A (high-divergence-apparent, always terminal-locked). Two causal routes converge on the same outcome:

- **A1 (usual):** Self-propelled— $F_i F_i$ -dominated, fed by vented extraction-heat once a coupled neighbor decouples, with $\text{Dpm} \rightarrow 0 \text{Dpm} \rightarrow 0$ convergence.
- **A2 (occasional):** Externally triggered—an external strike combines with a $\text{Dom} \rightarrow 0 \text{Dom} \rightarrow 0$ collision event, producing oscillation larger than self-propulsion alone could generate. The container careens back, finds no counter-oscillating partner, and the unopposed momentum forces the same terminal-locked outcome as A1.

Flip Condition B (low-divergence-apparent, externally driven, two distinct outcomes). Always $\sum_j \Delta_{ij} \sum_j \Delta_{ij}$ -dominated—driven by simultaneous multi-directional lateral knocks from neighboring high-divergence-apparent containers:

- **B1:** $\text{Dom} \rightarrow 0 \text{Dom} \rightarrow 0$ —meta converges with the visibly chaotic official layer, engaging only the smaller fluid mass (official and meta, the minority layers)—producing $180^\circ 180^\circ$, terminal-locked.
 - **B2:** $\text{Dpm} \rightarrow 0 \text{Dpm} \rightarrow 0$ —meta converges with the actually-calm operational layer, engaging the larger fluid mass (operational and meta, the majority layers)—completing a full $360^\circ 360^\circ$ transition.
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§10.5b.6: Vector-Aware Coupling

A single dominant external source imparts a force whose direction reverses over the source's oscillation cycle; summed over the tipping window, this is substantially self-cancelling and functions as a restoring, damping influence. Only simultaneous, multi-directional knocking—several high-divergence-apparent neighbors striking from divergent vectors at once—removes the restoring cancellation and produces a net constructive force:

$$\text{DriveB}(t) = \left| \sum_j \Delta_{ij}(t) \cdot u^j(t) \right| \quad \text{DriveB}(t) = \sum_j \Delta_{ij}(t) \cdot u^j(t)$$

where $u^j(t)u^j(t)$ is the unit direction of neighbor jj 's disturbance at time t .

§10.5b.7: Fluid-Mass Weighting

Define mass weights μ_o, μ_p, μ_m for the official, operational, and meta layers respectively (fitted per container, or adapter-supplied), with $\mu_p > \mu_o, \mu_p > \mu_o$ structurally in the low-divergence-apparent regime. The effective driving momentum for each B route is weighted by the mass of its converging pair:

$$M_{tip}(B1) = \int \mu_o \cdot \text{DriveB}(t) dt, M_{tip}(B2) = \int \mu_p \cdot \text{DriveB}(t) dt \quad M_{tip}(B1) = \int \mu_o \cdot \text{DriveB}(t) dt, M_{tip}(B2) = \int \mu_p \cdot \text{DriveB}(t) dt$$

Because $\mu_p > \mu_o, \mu_p > \mu_o$ in this regime, B2 is structurally more likely to cross into the reorganization range.

§10.5b.8: Coherence, Dissipation, and Flip Condition A's Momentum Functional

For Flip Condition A, some of the driving force is dissipated as internal friction, proportional to whatever divergence remains unresolved outside the converging pair.

Define the residual divergence at a convergence event as the pairwise divergence not involved in the convergence itself—for A1/A2's $D_{pm} \rightarrow 0$ or $D_{om} \rightarrow 0$, the residual is D_{op} —and a coherence factor:

$$\kappa(t) = 1 + D_{\text{residual}}(t), \kappa \in (0, 1] \quad \kappa(t) = 1 + D_{\text{residual}}(t), \kappa \in (0, 1]$$

with:

$$M_{\text{tip}}(A) = \int |F_i(t)| \cdot \kappa(t) dt \quad M_{\text{tip}}(A) = \int |F_i(t)| \cdot \kappa(t) dt$$

§10.5b.9: Outcome Thresholds

Two thresholds, $M_{\text{lock}} < M_c$ (both fitted), apply uniformly:

- **$M_{\text{tip}} < M_{\text{lock}} < M_c$** : No tipping event completes. The container self-arrests and returns to bounded oscillation.
 - **$M_{\text{lock}} \leq M_{\text{tip}} < M_c$** : Terminal-locked outcome per Definition 10.1—reached by A1, A2, or B1. Requires an external anchor to quiet residual oscillation.
 - **$M_{\text{tip}} \geq M_c$** : Full reorganization outcome—reached by B2 specifically, and structurally unreachable by any A route or by B1. $S(t)$ and $\delta R(t)$ reset toward zero; $dA(t)$ recovers toward dA_{max} .
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§10.5b.10: Prevention vs. Post-Flip Intervention

A low-divergence-apparent (medium-entropy) container's impending tipping event is comparatively visible in advance—its official layer is visibly chaotic ($\sigma_o(t)$ rising). Prevention—steadyng the container before it tips—is substantially safer than allowing the event and hoping for B2.

A high-divergence-apparent (high-entropy) container's impending tip is comparatively invisible in advance—its official layer presents as stable while the actual instability is hidden in the operational layer. Post-collapse intervention receives comparatively more emphasis for this regime in practice, not because prevention is less valuable in principle, but because it is harder to act on before the fact.

§10.5b.11: Steadyng vs. One-Time Reset

A low-divergence-apparent container that completes a B2 reorganization, or is steadied pre-flip, benefits from external anchoring primarily as a speed multiplier.

A high-divergence-apparent container is structurally different: merely righting or resetting it without addressing internal composition is insufficient, since its unaltered internal mechanics will trigger renewed violent oscillation almost immediately. This regime requires sustained steadying—continuous external containment disallowing both further oscillation and further extraction/displacement (§8.6a)—maintained over time, not applied once.

§10.5b.12: Combined Intervention

For both regimes, anchoring (mechanical/behavioral intervention, arresting oscillation and extraction) and active messaging change (external correction, or the internal rogue-metamessenger pathway) are more effective together than either alone. Anchoring buys the time messaging change needs to work without triggering premature confrontation.

§10.5b.13: Anchor Eligibility

- A low-entropy baseline container can safely anchor a destabilizing neighbor, but only by actively maintaining zero extraction and zero displacement flux for the full duration of the intervention— $X_{\text{anchor},i}(t)=0$ $X_{\text{anchor},i}(t)=0$ (§8.6a) throughout. Lapsing risks the anchor being drawn into an extraction cycle.
 - A semi-rigid horizontal field—a boundary configuration with no exploitable structure to extract from or displace onto—is structurally, not behaviorally, safe: $X=0$ $X=0$ by construction. This is the preferred anchor wherever available.
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§10.5b.14: Rogue Metamessengers and Slow-Correction Reversal

A small, coordinated subset of a high-divergence-apparent container's own meta-representation apparatus—aware the operational layer is corrupted and actively working against the official layer's beliefs—can raise $\Pi_f(t)$ $\Pi_f(t)$ deliberately, provided the rate of increase stays below whatever threshold triggers F_{conf} F_{conf} 's destabilizing surge. Formally: a controlled injection $d\Pi_f/dt=\epsilon_{\text{rogue}}$ $d\Pi_f/dt=\epsilon_{\text{rogue}}$, with ϵ_{rogue} ϵ_{rogue} small and bounded well under the rate that would cross Π_c Π_c abruptly. This permits D_{op} D_{op} to be metabolized incrementally through ordinary correction rather than forcing a confrontation event, allowing gradual regime transition—high-divergence-apparent toward low-divergence-apparent or, ideally, toward the low-entropy baseline—without ever triggering Flip Condition A.

§10.5b.15: Two Collapse-Layer Axioms

Axiom C1 (Truth-Only Reorganization).

The feedback operator FF (§6.4) admits stable, unforced fixed points only at convergence toward the container's true underlying signal. Convergence toward a false signal requires continuous external or self-generated forcing to be maintained, and decays toward the true-signal fixed point once that forcing is withdrawn.

Axiom C2 (Runaway Acceleration).

As a container approaches a hull-breach crossing (§10.5a) while remaining in Flip Condition A, its instantaneous divergence pressure accelerates rather than approaching the threshold at constant or decelerating rate. This predicts that $d(Dop)/dt$ and dS/dt should show a characteristic upward inflection in the window immediately preceding a hull-breach crossing.

Summary of Additions to §10.5b

Concept	Formalization	Coupling to Existing Content
Verticality operator	$V(t)=\Phi V(dh/dt, d^2h/dt^2, S, \delta R, \Pi f)$	High-entropy regime = $V(t) > \Theta V$
Horizontality operator	$W(t)=\Phi W(\Pi f, I, -dDop/dt, dR/dt)$	Low-entropy baseline = $W(t) > \Theta W$
90° Inversion	$I90:H \rightarrow CVI90:H \rightarrow CV$ (self-dual)	180° flip = one inversion; 360° flip = two inversions
Ultimate Convergence	$V(t) \rightarrow 0, W(t) \rightarrow 1, h(t) \rightarrow 0$	Recovery = return to horizontality; target of all anchors
Mass weighting and inversion	μ_0, μ_p, μ_m determine which inversion path occurs	B1 vs. B2 outcomes (§10.5b.7)

Horizontal
anchors

Boundary configuration with $X=0$ by
construction

Safe stabilization of
high-entropy
containers (§10.5b.11,
§8.6a)

10.6 Sources of Collapse

The collapse operator is generated by multiple coupled mechanisms: declining informational integrity, accumulated informational stress, resource depletion, recursive inconsistency, network fragmentation, maladaptive memory, excessive rigidity. No single variable is sufficient; collapse is fundamentally multicausal.

10.7 Progressive Collapse

$d(d_A)/dt < 0$ indicates progressive collapse. External failure becomes observable only after adaptive dimensionality falls below that required for organizational persistence — this distinguishes latent degradation from overt failure.

10.8 Recovery

Define the recovery operator U . Recovery satisfies $\dim(U(\mathcal{P})) > \dim(\mathcal{P})$. Recovery need not restore the original organization; adaptive systems frequently recover by constructing entirely new organizational trajectories.

10.9 Collapse Attractors

A collapse attractor $\mathcal{A}_C \subseteq \mathcal{A}$ is such that $X(t) \rightarrow \mathcal{A}_C$ for neighboring trajectories — institutional corruption, ecological desertification, chronic software instability, persistent organizational dysfunction, long-term physiological dysregulation. Once within a collapse attractor, ordinary adaptive feedback becomes insufficient for spontaneous recovery.

10.10 Critical Slowing Down

Let $\tau(X)$ denote the characteristic recovery time. $d\tau/d(d_A) < 0$ — recovery time increases as adaptive flexibility decreases. This provides a measurable early-warning indicator of impending collapse.

10.11 Recursive Collapse

Collapse propagates across recursive levels via the recursive coupling operators of Part 5. Local collapse may remain localized, or propagate upward, or propagate downward, depending upon recursive coupling strength.

10.12 Collapse as Loss of Adaptive Possibility

Collapse is defined as the progressive contraction of the space of feasible adaptive transformations available to a recursively organized information-processing system.

The Collapse Principle. Collapse is the geometric contraction of adaptive possibility. Stability is the preservation of adaptive possibility. Recovery is the expansion of adaptive possibility.

Transition to Part 11

The next section derives the principal observables of RICD as measurable functionals arising from the underlying mathematics.

Part 11. Derived Observables

Observables are functionals defined on the adaptive state manifold, not primitive variables. Different applications may employ different empirical measurements while referring to the same underlying mathematical object.

11.1 Observation Functionals

$O : \mathcal{A} \rightarrow \mathbb{R}$, or more generally $O : \mathcal{A} \rightarrow \mathbb{R}^m$.

11.2 Informational Integrity

$I = G(D_{op}, D_{om}, D_{pm})$, G monotonically decreasing in each divergence. High informational integrity indicates coherent internal organization.

11.3 Feedback Permeability

$\Pi_f \in [0,1]$. Empirically estimated from learning rates, correction latency, implementation success, error-resolution frequency, adaptive responsiveness.

11.4 Informational Stress

$S(t) = \int_0^t K(t, s) (1 - \Pi_f(s)) D(s) ds$ — inferred from the interaction between divergence, memory, and permeability. Persistent high stress predicts increasing adaptive rigidity even before overt failure becomes visible.

11.5 Adaptive Capacity

$C_A = \phi(d_A)$, $\phi' > 0$. Measures the size of the feasible adaptive region rather than instantaneous performance — distinguishing currently-functioning-but-inflexible systems from genuinely resilient ones.

11.6 Adaptive Rigidity

$R_g = 1 - \tilde{C}_A$, where $\tilde{C}_A$ is a normalized adaptive capacity.

11.7 Fragmentation

$F = \sum_k [w_k D(P^{(k)}, P^{(k+1)})]$, w_k recursion-level weights. High fragmentation indicates poor consistency between neighboring organizational scales — organizational silos, conflicting policies, software layer incompatibilities, ecological decoupling.

11.8 Network Coherence

$N = \Psi(G, W, P)$. High values correspond to efficient information propagation, robust resource exchange, resilient connectivity.

11.9 Resource Sustainability

$s_R = \limsup_{T \rightarrow \infty} \{ (1/T) \int_0^T (P_{\text{prod}} - W) dt \}$. Positive values indicate sustainable adaptation; negative, progressive resource debt.

11.10 Collapse Potential

$P_C = \Theta(I, S, R, d_A, N)$, Θ monotonically increasing in adaptive degradation. Varies continuously across the adaptive manifold, estimating systemic vulnerability rather than declaring failure.

Computational instantiation. This is RICD's direct generalization of the companion FDFM document's $D = \text{sigmoid}(\phi(S, M, F))$ — same link function, same “weighted-linear-combination-through-a-squashing-function” structure, extended from three inputs to the five RICD already tracks:

$$P_C(t) = 1 / (1 + \exp[-(\kappa_1(1-I(t)) + \kappa_2\tilde{S}(t) + \kappa_3(1-\Pi_f(t)) + \kappa_4(1-d_A(t)/d_A^{\text{max}}) + \kappa_5(1-N(t))))]),$$

with $\kappa_1 \cdots \kappa_5 > 0$ fitted weights and each bracketed term already defined and normalized to $[0,1]$ elsewhere in this document (I : §6.6; $\tilde{S}$: §6.7/§10.3; Π_f : §6.5; d_A : §10.3; N : §8.9). $P_C \in (0,1)$ throughout, increasing in every source of degradation as required, and — unlike the companion document's D , which was shown to have been redefined inconsistently between its theoretical and empirical sections (Part 15 addresses this directly) — P_C has exactly one definition, used identically whether computed from a Bayesian posterior or a forward simulation.

11.11 Observable Invariance

If two systems satisfy $X_1 < X_2$ with respect to a particular adaptive property, any admissible observation procedure should preserve that ordering up to measurement error.

The Observable Principle. Observable quantities are functionals defined on the underlying adaptive state manifold, revealing informational organization, recursive structure, resource sustainability, and adaptive possibility in empirically measurable form.

Transition to Part 12

The next section develops the first formal mathematical results of RICD.

Part 12. Fundamental Mathematical Results

Unless otherwise stated, all statements assume that the informational dynamics, resource functionals, and recursive operators satisfy the regularity conditions introduced in the preceding sections, including the scope conditions on the information manifold specified in §4.1.

Theorem 12.1 (Existence of Adaptive Trajectories). Let $X(t) = (P, R, G, M)$ denote the adaptive state of a container. Suppose the governing evolution operators are continuous and locally Lipschitz on the adaptive state manifold. Then, for every admissible initial condition $X(0) = X_0$, there exists a unique local adaptive trajectory $X(t)$ defined on some interval $[0, T)$.

Sketch of proof. The coupled evolution equations define an ordinary (or hybrid) dynamical system on the adaptive manifold. Under local Lipschitz continuity, the Picard–Lindelöf theorem guarantees local existence and uniqueness. Recursive coupling does not alter this conclusion provided the aggregation and projection operators remain continuous. ■

RICD therefore defines a mathematically well-posed dynamical system rather than a collection of disconnected variables — this is the one result in this section that follows directly from a named, standard theorem under stated hypotheses.

Theorem 12.2 (Monotonicity of Informational Stress). Suppose $K(t, s) \geq 0$, $D(s) \geq 0$, and $0 \leq \Pi_f(s) \leq 1$. Then $S(t) = \int_0^t K(t, s)(1 - \Pi_f(s)) D(s) ds$ is nonnegative. Furthermore, if unresolved informational divergence persists, $S(t)$ is nondecreasing.

Sketch of proof. Every factor inside the integral is nonnegative, so the integral is nonnegative; additional unresolved divergence contributes nonnegative area to the accumulated stress functional. ■

Stress cannot decrease simply because time passes — it decreases only through genuine reduction of unresolved informational discrepancy or through explicit forgetting mechanisms incorporated into the memory kernel.

Proposition 12.3 (Bounded Informational Integrity). Suppose G is normalized. Then $0 \leq I \leq 1$, with $I = 1$ if and only if all representation divergences vanish. Perfect informational integrity corresponds to complete consistency among internal representations; it does not imply optimal adaptation or maximal performance.

Theorem 12.4 (Resource Feasibility, conditional form). Suppose $R(0) \geq 0$, and suppose further that informational work is self-limiting in the sense that $W(t) \leq w(R(t))$ for some function w with $w(0) = 0$ and w continuous — i.e., the container cannot perform work it lacks the resources to fund, rather than merely being assumed not to (this is the corrected

hypothesis; the previous draft assumed the conclusion). If $P_{\text{prod}} + E_{\text{exch}} \geq 0$ whenever R is at or near zero, then $R(t) \geq 0$ for all admissible times. **This hypothesis is no longer merely hypothetical:** §7.3's concrete instantiation $W(t) = \min[W_{\text{raw}}(t), R(t)]$ satisfies $w(R) = R$ exactly, so any implementation using that clipping rule inherits this theorem's guarantee automatically rather than needing to verify the hypothesis separately per application.

Sketch of proof. Consider the boundary $R = 0$. There, $w(0) = 0$, so $W \leq 0$ forces $W = 0$ by nonnegativity of work (Axiom 3 gives strict positivity only for nontrivial updates; at the boundary no work can be funded). Hence $dR/dt = P_{\text{prod}} - 0 + E_{\text{exch}} \geq 0$ by hypothesis, so R cannot be driven below zero from $R = 0$: the boundary is repelling or at worst reflecting. By continuity, $R(t) \geq 0$ follows on any interval where the hypotheses hold. ■

This is a genuine, if narrower, feasibility result: nonnegativity is derived from a stated mechanism (work is bounded by available resource) rather than assumed as the hypothesis that also serves as the conclusion. An application that cannot justify $W \leq w(R)$ with $w(0) = 0$ has not established resource feasibility and should not cite this theorem.

Proposition 12.5 (Collapse Implies Reduced Adaptive Dimensionality). If $\mathcal{C}: \mathcal{S} \rightarrow \mathcal{S}$ is the collapse operator, then $\dim(\mathcal{S}) < \dim(\mathcal{S})$, by Definition 10.1 directly. This is definitional rather than a further derived result, and is recorded here as a proposition rather than a theorem for that reason.

Theorem 12.6 (Recovery Expands Adaptive Possibility). If recovery succeeds, $\dim(U(\mathcal{S})) > \dim(\mathcal{S})$, again following from the definition of U in §10.8. Recovery is mathematically distinct from returning to a previous state; adaptive systems frequently recover through novel organizational structures rather than exact restoration.

Proposition 12.7 (Recursive Consistency). If every aggregation operator preserves informational compatibility, then recursive inconsistency $F = 0$ if and only if every recursion level possesses identical informational organization after projection. Perfect recursive consistency is a limiting case; real adaptive systems generally possess nonzero but bounded recursive inconsistency.

Proposition 12.8 (Network Fragmentation and Adaptive Capacity — restored to a derivable proposition). Two drafts ago this was stated as a theorem with no supporting coupling; one draft ago it was downgraded to a conjecture for exactly that reason. §10.3's concrete instantiation now supplies the missing coupling directly: $d_A(t) = d_A^{\max} \cdot \exp[-\beta_S \tilde{S} - \beta_R \delta \tilde{R} - \beta_F \tilde{F}]$, where $\tilde{F}$ is normalized fragmentation and, per §11.7/§8.9, network coherence N feeds into fragmentation (lower N contributes to higher measured F between containers whose coordination has degraded). Consequently:

$$\partial d_A / \partial \tilde{F} = -\beta_F \cdot d_A < 0 \text{ for } \beta_F > 0, d_A > 0,$$

so $d\tilde{F}/dt > 0$ (worsening fragmentation, which follows from $dN/dt < 0$ under §8.9's coherence definition) implies $d(d_A)/dt < 0$ directly, by the chain rule, given the explicit functional form above. This is now a genuine derivation from a stated equation rather than a restated hypothesis, conditional only on accepting §10.3's exponential form for d_A as the

operative definition for a given application — which is the same sense in which Theorem 12.4 is conditional on accepting §7.3's clipped W . Applications using a different (independently justified) d_A functional would need to re-derive this coupling for that functional; the theorem's status is tied to the default instantiation, not proven for every conceivable one.

Corollary 12.9 (Positive Feedback Instability). If informational divergence increases informational work, resource depletion reduces adaptive updating, and reduced adaptive updating increases divergence, then the coupled system admits positive-feedback regimes in which $dD/dt > 0$ and $dR/dt < 0$ simultaneously reinforce one another. This does not imply inevitable collapse; external intervention, resource infusion, or successful informational correction may interrupt the cycle.

Theorem 12.10 (Scale Invariance of the Formalism). If the recursive operators satisfy the axioms of Part 5, the governing equations retain their form under changes of recursion level; only the interpretation of the variables changes. The same mathematical framework therefore applies to cells, organisms, organizations, institutions, ecosystems, and software systems without modification of the governing formalism.

The General Persistence Criterion

Criterion 12.1. A recursively organized adaptive system remains viable provided:

1. informational integrity remains bounded away from zero;
2. adaptive resources remain nonnegative and sustainable;
3. adaptive degrees of freedom do not contract below the minimum required for continued organization;
4. recursive coupling preserves sufficient coherence across scales;
5. network connectivity remains capable of supporting informational and resource exchange.

Failure of one criterion need not imply collapse. Persistent degradation across several criteria, however, contracts adaptive possibility and drives the system toward collapse.

The Fundamental Result of RICD. Adaptive persistence emerges from the coupled evolution of informational organization, resource sustainability, recursive structure, network connectivity, and historical memory. Collapse corresponds to the progressive contraction of adaptive possibility produced by failures across these interacting domains.

Transition to Part 13

The next section situates RICD within the broader landscape of dynamical systems, information theory, network science, control theory, cybernetics, statistical mechanics, and complexity science.

Part 13. Relationship to Existing Mathematical Frameworks

13.1 Relation to Classical Dynamical Systems

RICD enlarges the notion of state to $X = (P, R, G, M)$, incorporating informational organization, resource dynamics, adaptive networks, and historical memory. Classical dynamical systems generally assume fixed state variables; RICD allows both the variables and the organizational architecture through which they interact to evolve recursively.

13.2 Relation to Information Geometry

RICD adopts the geometric perspective of information geometry as its informational foundation, subject to the scope conditions of §4.1, and departs from it by coupling the informational manifold to recursive organization, adaptive resources, and evolving interaction networks, and by making the geometry itself history-dependent (§9.10). Traditional information geometry studies geometry on probability spaces; RICD studies adaptive systems whose geometry itself evolves.

13.3 Relation to Network Science

RICD treats network topology as an emergent adaptive quantity rather than a fixed modeling assumption. Conventional network models typically evolve node states on a graph; RICD simultaneously evolves both the node states and the graph itself.

13.4 Relation to Control Theory

FDFM extends feedback beyond an external controller to an intrinsic property of every adaptive container, with feedback effectiveness depending upon informational integrity, permeability, memory, and available resources rather than controller design alone.

13.5 Relation to Cybernetics

RICD may be viewed as a mathematical continuation of the cybernetic tradition, developing communication, adaptation, recursive organization, and self-regulation within a formal framework based on information geometry, recursive operators, adaptive resources, and evolving networks.

13.6 Relation to Statistical Mechanics

The recursive aggregation operators of Part 5 permit genuine emergence rather than simple averaging — higher-order containers acquire new informational states, resource dynamics, and boundary operators not generally reducible to microscopic descriptions alone.

13.7 Relation to Complexity Science

RICD's principal contribution is a common mathematical language for recursive hierarchy, adaptive networks, informational organization, resource economics, historical memory, collapse, and recovery, derived from a common set of primitive objects and axioms rather than independent models per phenomenon.

13.8 Relation to Machine Learning

Learning becomes one example of informational transformation within an adaptive container. RICD additionally models resource constraints, recursive organizational structure, memory, network evolution, and adaptive collapse, which often remain external to conventional optimization frameworks.

13.9 Relation to Organizational Theory

RICD interprets quantities such as trust, bureaucracy, communication, fragmentation, and resilience as observables derived from the underlying adaptive state, rather than independent empirical variables — allowing different organizations to employ different measurements while remaining mathematically comparable.

13.10 A Unified Perspective

Conceptual Contributions.

1. Recursive containers provide a scale-independent primitive for adaptive organization.
2. FDFM formalizes internal informational consistency as a derived process operating within each container.
3. Adaptive possibility replaces threshold-based failure as the primary mathematical characterization of collapse.
4. History-dependent information geometry allows accumulated experience to alter future adaptive trajectories.
5. Emergent network topology replaces fixed interaction graphs.
6. Resource expenditure is derived from informational transformation rather than introduced as an independent state variable.
7. Observables are derived from the adaptive state manifold rather than postulated independently.

The Integration Principle. RICD does not replace existing mathematical theories of adaptive systems; it provides a common recursive framework within which their complementary strengths can be combined to analyze the organization, evolution, persistence, and collapse of complex adaptive systems across scales.

Transition to Part 14

The final section synthesizes the framework and outlines future directions.

Part 14. Discussion, Limitations, and Future Directions

14. Discussion

RICD was developed from the premise that adaptive systems across widely differing domains share a common organizational structure. The theory introduces a small set of primitive mathematical objects — information states, containers, boundaries, resources, memory, and recursive organization — from which informational integrity, fragmentation, resilience, rigidity, and collapse are subsequently derived.

Collapse has been reformulated as the progressive contraction of adaptive possibility; recovery, as its restoration or expansion.

14.1 Principal Contributions

First, recursive containers as scale-independent mathematical primitives. Second, a geometric description of informational organization based upon probability manifolds. Third, FDFM formalized as a derived representation of internal informational dynamics. Fourth, a unified treatment of adaptive resources, memory, recursive organization, and evolving interaction networks. Fifth, a geometric definition of adaptive collapse based upon contraction of feasible adaptive transformations.

14.2 Limitations

The present formulation is a fully computable mathematical framework, not a complete theory in the sense of having settled every open mathematical question, and several limitations should be stated precisely:

- The information metric g is defined only under the scope conditions given in §4.1, and Definition 4.2's six candidate distance functionals are not mutually substitutable once curvature or geodesics are invoked (§4.2) — an application needing those constructions must commit to Fisher–Rao. The default computational instantiations throughout Parts 6–11, however, use only the scalar (non-Riemannian) layer, per §6.3's JS-divergence default, so a collapse tracker built from this edition does not need to resolve the metric-choice question at all unless it separately wants geometric early-warning indicators.
- Every quantity that was previously an unspecified functional now has a stated default form (Part 15 consolidates all of them), but every default form carries fitted coefficients whose values are not fixed by the theory. The theory constrains the *shape* of these relationships (monotonicity, boundedness, the specific couplings derived in Part 12); it does not and should not predict the numeric coefficients for any particular system in advance of fitting them to data.
- The framework establishes local well-posedness (Theorem 12.1) under standard regularity assumptions but does not yet include global existence, bifurcation structure, or asymptotic stability analysis for the nonlinear coupled system that results once every Part 6–11 instantiation is composed together.
- Theorem 12.4 and Proposition 12.8 are conditional on accepting this edition's default instantiations (§7.3's clipped W ; §10.3's exponential d_A) as the operative

definitions. An application substituting different concrete forms for W or d_A would need to re-verify or re-derive these results for its own choices rather than inherit them automatically.

- The commensuration step in §5.2 (mapping heterogeneous child Ω 's onto a shared severity scale before aggregation) is, by design, left to the domain adapter. Nothing in the engine can verify that a given σ_i is a reasonable choice — that judgment is inherently substantive and domain-specific, and mis-specifying it would silently distort every downstream quantity computed on C^* .

These limitations mark exactly where domain judgment must enter — Part 15's adapter contract exists to isolate that judgment to one well-defined interface rather than let it leak into the engine.

14.3 Computational Development

A numerical simulation environment for recursive adaptive systems under varying assumptions regarding informational geometry, memory kernels, resource constraints, and adaptive network topology would allow systematic investigation of collapse attractors, critical slowing down, adaptive recovery, recursive propagation of disturbance, organizational phase transitions, and multiscale resilience. Because the governing formalism is domain-independent, a common computational engine may be adapted to diverse applications through alternative observation operators and parameterizations.

14.4 Empirical Validation

Testable predictions include: increasing informational stress should precede observable collapse; declining adaptive degrees of freedom should occur before catastrophic failure; persistent reductions in feedback permeability should predict growing rigidity; recursive fragmentation should reduce adaptive capacity across organizational scales; prolonged resource debt should amplify informational divergence through positive feedback. These may be evaluated within biological systems, software engineering, organizational management, ecological monitoring, public administration, healthcare, education, and other adaptive environments.

14.5 Future Mathematical Development

Open directions include: global existence and uniqueness under broader conditions; invariant manifolds and adaptive attractors; bifurcation theory for recursive adaptive systems; stochastic perturbations on evolving information manifolds; extension of recursive operators to hypergraphs and higher-order simplicial complexes; information-geometric variational principles; renormalization across recursive organizational scales; and canonical metrics for adaptive possibility. Proposition 12.8's coupling is now specified (§10.3), so the remaining open item there is empirical rather than mathematical: testing whether the exponential $d_A(\tilde{F})$ form is the right one for a given domain, or whether fitted data favor a different functional relationship between fragmentation and adaptive capacity.

14.6 Broader Research Program

Potential application domains include software reliability and cybersecurity, artificial intelligence, robotics, organizational governance, institutional analysis, healthcare systems, education, ecological resilience, infrastructure management, economic systems, disaster response, and evolutionary biology — representing adaptive organization within a common recursive mathematical language while preserving domain-specific observables.

14.7 Concluding Remarks

RICD proposes that the diversity of adaptive systems observed across nature, technology, and society may be understood through a shared mathematical architecture based upon recursive information-processing containers. Information, resources, memory, boundaries, and networks are interacting components of a unified adaptive geometry. Stability emerges through the preservation of adaptive possibility; collapse, through its contraction; recovery, through its expansion.

The present work should be understood not as the culmination of a theory but as the beginning of an extensible mathematical program for the study of adaptive systems — one whose internal consistency, per this revision, now rests on stated rather than assumed conditions throughout, and whose every previously-abstract functional now has a concrete default form suitable for direct computational implementation (Part 15).

Part 15. Computational Specification

Parts 1–14 establish RICD as a mathematical theory. This part exists for a different purpose: to consolidate everything needed to implement it, so that building a working collapse tracker is a matter of supplying data to a fixed interface rather than re-deriving formulas from the preceding text. Everything in this part restates definitions given earlier; nothing here is new theory.

15.1 The Two-Layer Architecture

Every RICD implementation separates into exactly two layers, and the boundary between them is the boundary between domain judgment and fixed mathematics:

The domain adapter (written once per system studied) supplies only raw primitives — it never computes a derived quantity itself.

The engine (written once, reused across every domain) consumes only what the adapter supplies and computes every derived quantity by the fixed formulas below. The engine has no domain-specific code in it anywhere; if it needs to “know” it’s looking at a justice system versus a cell versus a corporation, the adapter boundary has been drawn in the wrong place.

15.2 Adapter Contract — What Every Domain Implementation Must Supply

For each container C in the system being modeled:

1. Ω — the outcome space, stated explicitly (discrete category set is the default assumption throughout this part; continuous Ω requires substituting a continuous-compatible D per §4.2).
2. O_o, O_p, O_m — three data-extraction functions returning empirical distributions over Ω at time t: the official representation (codified rule, stated policy, formal specification), the operational representation (measured actual behavior/outcomes), and the meta-representation (independent oversight, audit, or third-party review — §6.1's requirement that this be a genuinely independent channel, not derived from the O-P gap, is binding here).
3. $R(t)$ — a resource observation, in whatever unit is natural to the container, plus $R^{\max}$ for that container (used in §5.6's normalization and §10.4's rigidity).
4. $P_{\text{prod}}(t), E_{\text{exch}}(t)$ — observed production and net boundary exchange, feeding §7.4 directly.
5. **Raw permeability signals** — counts of corrective actions and error signals, and a response-lag measurement, feeding §6.5's Π_f proxy.
6. $d_A^{\max}$ — a domain-specified ceiling on adaptive capacity (§10.3), and, optionally, a direct feasible-transformation count if the domain supports one (overriding the default exponential-decay d_A).
7. For containers that are children of a larger aggregate: σ_i , the commensuration map from that container's Ω onto the parent's shared severity scale (§5.2).

Nothing else is domain-specific. Every quantity below is computed by the engine from these seven inputs, identically regardless of what C represents.

Warning: apparent volatility is not a stability proxy, and using it as one inverts the risk ranking exactly where it matters most. §10.5b's entropy regime classification exists because low apparent variance in the cheaply-observed layer (σ_o , the official/declared channel — usually the easiest and cheapest data to pull for a real domain: published statistics, stated policy, official reporting) is the *signature of the high-divergence-apparent regime*, not evidence of health. An adapter that observes only O_o and treats its low volatility as a stability signal will systematically rank the most dangerous containers in a system as its safest ones, and the genuinely recovering containers (low-divergence-apparent, visibly volatile in O_o while actually stable in O_p) as its most dangerous. This is not a minor calibration risk; it is the specific failure mode the regime classification was built to prevent, and every adapter implementation must pull both O_o and O_p , at comparable observation quality, or the engine's regime classification cannot function as intended.

15.3 Engine Reference — Every Derived Quantity, One Formula Each

Symbol	Meaning	Formula	Defined in	Fitted parameters
D_{ij}	pairwise divergence	smoothed Jensen–Shannon	§6.3	—
Π_f	feedback permeability	$\text{logistic}(\kappa_{\Pi} \cdot [\text{corrective/error} - \text{lag}])$	§6.5	κ_{Π}
I	informational integrity	$\exp[-\alpha \cdot \Sigma D_{ij}]$	§6.6	α
$S(t)$	informational stress	recursive exponential-kernel update	§6.7	λ
$W(t)$	informational work	$\min[\kappa_W \cdot D^p, R(t)]$	§7.3	κ_W, p
Γ_R	resource recovery	$\rho \cdot (R_{\text{target}} - R)$	§7.7	ρ, R_{target}
$\delta R(t)$	resource debt	$\int (W - P_{\text{prod}} - E_{\text{exch}}) ds$	§7.9	—
A	aggregation (parent state)	commensurate then resource-weight-pool	§5.2	—
Ξ	edge existence	$\text{freq} \times \min(\Pi_{f,i}, \Pi_{f,j})$ vs θ	§8.3	θ
Λ_{ij}	network coupling	$\gamma_{\text{net}} \cdot W_{ij} \cdot \Pi_{f,i}$	§8.5	γ_{net}
$W_{ij}(t)$	edge weight update	$\beta_{\text{net}} \cdot \text{success} - \gamma_{\text{net}} \cdot \text{ignored}$	§8.7	$\beta_{\text{net}}, \gamma_{\text{net}}$
R_N, N	network resilience/coherence	$\lambda_2(\text{Laplacian}) \times \text{mean}(\Pi_f)$, normalized	§8.9	—
$d_A(t)$	adaptive degrees of freedom	$d_A^{\text{max}} \cdot \exp[-\beta_S \tilde{S} - \beta_R \delta \tilde{R} - \beta_F \tilde{F}]$	§10.3	$\beta_S, \beta_R, \beta_F$
$d_A(C^*)$	recursive d_A aggregation	$\Sigma w_i \cdot d_A(C_i) \cdot (1 - F(C^*))$	§10.3	—

Symbol	Meaning	Formula	Defined in	Fitted parameters
R_g	adaptive rigidity	$1 - d_A/d_A^{\max}$	§10.4	—
Hull breach	collapse-onset signal	$(1-I) + \tilde{S} > \theta_{\text{hull}} \cdot \Pi_f$	§10.5a	θ_{hull}
σ_o, σ_p	entropy regime classification	trailing-window variance of P_o, P_p	§10.5b	window length
F_{conf}	confrontation trigger	$\zeta \cdot \Pi_f \cdot D_{\text{op}}$, active near Π_c crossing	§10.5b	ζ, Π_c
$\kappa(t)$	coherence factor	$1/(1+D_{\text{residual}})$	§10.5b	—
M_{tip}	tipping-momentum functional	$\int$	Drive	$\cdot \kappa dt$, Drive = F_i or $\Sigma \Lambda_{ij}$
Outcome	tipping-event resolution	$A1/A2 \rightarrow \text{lock}$; $B1(\mu_o) \rightarrow \text{lock}$; $B2(\mu_p) \rightarrow \text{reorganize}$, by M_{tip} vs M_{lock}, M_c	§10.5b	M_{lock}, M_c
μ_o, μ_p, μ_m	representation-layer mass weights	fitted or adapter-supplied; $\mu_p > \mu_o$ in low-div-apparent regime	§10.5b	μ_o, μ_p, μ_m
$\rho(j)$	coupling persistence	high for low-divergence-apparent, low for baseline	§8.6a	—
$X_{\{i,j\}}$	extraction flux	$\chi \cdot \Lambda_{ij} \cdot \rho(j) \cdot R_j$	§8.6a	χ
H_i	vented metabolic-failure heat	$(1-\eta_i) \cdot X_{\{i,j\}}$	§8.6a	η_i

Symbol	Meaning	Formula	Defined in	Fitted parameters
$V(t)$	verticality-signal (genesis branching)	shock-transmitted or ambient grid field	§10.5b	$v_threshold$
ε_rogue	slow-correction injection rate	$d\Pi_f/dt$, held below F_conf -triggering rate	§10.5b	ε_rogue
F	fragmentation	$\Sigma w_k \cdot D(P^{(k),P}(k+1))$	§11.7	w_k
$P_C(t)$	collapse potential	$\text{sigmoid}(\kappa_1(1-I) + \kappa_2 S + \kappa_3(1-\Pi_f) + \kappa_4(1-d_A/d_A^m ax) + \kappa_5(1-N))$	§11.10	$\kappa_1 \cdots \kappa_5$

Every entry in the rightmost column is a free parameter analogous to a, b, c, d, e, r in the companion FDFM document — not guessed, not fixed by the theory, and exactly what the Bayesian state-space layer exists to fit once real data and a chosen system are in hand.

15.4 What Happens Next (Deliberately Not Done in This Document)

This document stops short of choosing Ω , writing the adapter for any specific system, or fitting any of the coefficients in §15.3 to real data — that work is domain-specific by construction (§15.1) and, per the scope of this revision, comes after the mathematics is settled, not before. The next step is choosing a system, defining its adapter (§15.2) concretely, and building the Bayesian state-space model that infers §15.3's fitted parameters from that system's data — mirroring the companion FDFM document's PyMC layer, generalized to RICD's larger parameter set and corrected so that every latent variable (I, S, d_A, P_C) is computed by exactly one formula throughout, used identically in the generative model, the inference likelihood, and any downstream simulation — unlike the companion document's D , which was defined one way in the SDE/Bayesian layer and a different, disconnected way in the empirical feature-engineering layer.

Final Statement

Recursive Information-Container Dynamics advances the proposition that adaptive systems, despite their immense diversity, can be described within a common mathematical framework in which recursively organized information-processing containers evolve through the coupled dynamics of information, resources, memory, boundaries, and networks. Within this framework, persistence is the preservation of adaptive possibility, collapse is its contraction, and recovery is its expansion. Every quantity in that framework, as of this edition, can be computed from real data through a stated formula — the theory and the tool are now the same object.